

# Supplementary material (Online Resource 1): the derivations to the article “Universe24: one integer rule on a cubic lattice, read through clicks, gives the wave equation, the quantum of action and a weak-field metric as three readings of one line”, submitted to Foundations of Physics

Alon Gonen

Independent researcher, Haifa, Israel

alon@defounder.ai    ORCID 0009-0008-2701-4599

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This supplement gives the algebraic steps of every result the paper marks DERIVED or COMPUTED without a proof in its text, in the order of the paper’s sections; the theorems proved in the paper are not repeated. Each derivation names its inputs from the paper, its steps, and, where a number was computed, the computation and its check. The notation is the paper’s: Rule3’s coefficients at a Node of clock pace  $p_0$  and Link paces  $p_a$  are  $R_a = 2 \text{ num } p_a^2$  on the two arrivals of the axis  $a$ ,  $S = 12 \text{ den } \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{ num } \sum_a p_a^2$  on the Node’s own level, and the wall  $w = 6 \text{ den } \Gamma^2$ ; the content read at the Node is  $x = c/\Gamma$ , also written  $U$ ;  $g = (\text{den} - \text{num})/\text{den} = 1 - \cos \omega_0$ .

## The path from Rule3 to each result

Every result of the paper is reached from Eq. (1), Rule3’s line with its one write and its three readings, by a chain of the derivations below; where a link of a chain is an assumption of the model, it is named in the chain, so that each path is closed in the algebra with its assumptions explicit.

- *Light’s speed and dispersion*: Eq. (1) at the vacuum’s pace  $\rightarrow$  the band, S.1.
- *Gravity as a slowed clock, light’s index, the bending and Shapiro’s delay*: Eq. (1)  $\rightarrow$  the paces, Eq. (3) (the assumption: the level read once into the clock and twice into the Link)  $\rightarrow$  the band at a pace, S.1  $\rightarrow$  the index, S.2  $\rightarrow$  the ray integral, Eq. (15) and S.20.
- *The redshift of a bound rotation*: Eq. (1)  $\rightarrow$  S.1 at  $k = 0$  with the clock’s pace  $\rightarrow$  S.15.
- *The fall, the Newtonian potential the orbits measure and the post-Newtonian parameters*: Eq. (1)  $\rightarrow$  S.1 at small  $k$   $\rightarrow$  the inertia and the ray equations, S.18  $\rightarrow$  S.19.
- *Lorentz’s factor to the second order*: Eq. (1)  $\rightarrow$  the exact band of a free record, S.1  $\rightarrow$  the rotation at the moving centre, S.21, closed in the algebra for a free quantum of any family; for a body, the body’s own band from its fixed point, S.13, computed by the generator, enters the same S.21.
- *De Broglie’s relation*: Eq. (1)  $\rightarrow$  S.1  $\rightarrow$  S.22.
- *From the Node to us, bodies, clusters and the observer*: Eq. (1) with the write  $\rightarrow$  the fixed point and the functional, S.13  $\rightarrow$  the stability, Theorem 4  $\rightarrow$  the cloud and the free fall (computed)  $\rightarrow$  the clicks between bodies, S.9, S.14 to S.17  $\rightarrow$  the clusters and the observer as a cluster (Section 7.5, with the assumption that the universe is bound).
- *The count*: Eq. (1)  $\rightarrow$  the conserved form, Theorem 3  $\rightarrow$  the share, S.5 and S.6  $\rightarrow$  the invariant and the drift, S.14.
- *Light born by the write, absorption and emission*: Eq. (1)’s write line  $\rightarrow$  the Wronskian, S.9  $\rightarrow$  the resonance, S.16.
- *Bell*: Eq. (1) linear  $\rightarrow$  the parts’ determinism (Section 9.4)  $\rightarrow$  the declared credit (the assumption)  $\rightarrow$  S.10.
- *The two slits*: Eq. (1) stepped in real arithmetic on the declared world  $\rightarrow$  S.23 and the computed numbers of Section 9.
- *The constants and the bounds*: Eq. (1)  $\rightarrow$  S.1  $\rightarrow$  S.24 (the Link); the field’s rest  $\rightarrow$  S.25 (Newton’s constant); S.11 (the reach), S.12 (the two speeds).

## The model (Section 3 of the paper)

**Derivation S.1** (the band at a pace, Eq. (4) of the paper). Inputs: *Rule3’s line at the vacuum’s paces, where the two Links of an axis share one  $R_a = 2 \text{ num } \Gamma^2$ ,  $w a_{\text{next}} = \sum_a R_a (\text{arr}_{+a} + \text{arr}_{-a}) + S a_{\text{now}} - w a_{\text{before}}$ , with the remainders dropped (they are bounded by  $w$  and do not move a plane wave’s rotation).* Steps: *for a plane wave*

$a_t(\mathbf{x}) = A \cos(\mathbf{k} \cdot \mathbf{x} - \omega t)$  the two arrivals of the axis  $a$  sum to  $2 \cos k_a a_{\text{now}}$  and  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$ , so

$$2 \cos \omega = \frac{2 \sum_a R_a \cos k_a + S}{w} = \frac{4 \text{num} \sum_a p_a^2 \cos k_a + 12 \text{den} \Gamma^2 - 12(\text{den} - \text{num}) p_0^2 - 4 \text{num} \sum_a p_a^2}{6 \text{den} \Gamma^2},$$

that is  $\cos \omega = 1 - g(p_0/\Gamma)^2 - \frac{\text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2 (1 - \cos k_a)$ , which is Eq. (4). At the vacuum's pace,  $p_0 = p_a = \Gamma$ , this is  $\cos \omega = \text{num}/\text{den} - \frac{\text{num}}{3 \text{den}} \sum_a (1 - \cos k_a) = \frac{\text{num}}{3 \text{den}} \sum_a \cos k_a$ ; at  $k = 0$ ,  $\cos \omega_0 = \text{num}/\text{den}$ . Light's speed and dispersion: for  $[1, 1]$  along one axis  $\cos \omega = (2 + \cos k)/3$ ; expanding,  $1 - \omega^2/2 + \omega^4/24 = 1 - k^2/6 + k^4/72$ , so  $\omega = (k/\sqrt{3})(1 - k^2/36 + \dots)$  and the group velocity  $d\omega/dk = \sin k/(3 \sin \omega) = (1/\sqrt{3})(1 - k^2/12 + \dots)$ : the speed  $1/\sqrt{3}$  at long wavelength and the group speed falling as  $1 - 0.083 k^2$ . Check: at  $k = 0.05, 0.1$  and  $0.2$  the exact  $\sqrt{3} d\omega/dk - 1$  over  $k^2$  is  $-0.0833, -0.0834$  and  $-0.0835$ .

**Derivation S.2** (light's index at a content, Section 3.4). Inputs: Eq. (4) with the paces of Eq. (3),  $p_0 = \Gamma(1 - 1/\Gamma)^L = \Gamma N$ ,  $N = e^{-x} + O(1/\Gamma)$  at the content  $x = L/\Gamma$ , and  $p_a = p_0^2/\Gamma = \Gamma N^2$  for a reader at the weight 1 and no tension. Steps: for light,  $g = 0$ , so  $\cos \omega = 1 - \frac{1}{3} N^4 \sum_a (1 - \cos k_a)$ , and at long wavelength  $\omega = N^2 |\mathbf{k}|/\sqrt{3}$ . Light's speed at the content  $x$  is  $N^2 = e^{-2x}$  times its vacuum speed, so its index is  $n = e^{2x} = 1/(1 - 2x) + O(x^2)$  and  $n - 1 = 2x + O(x^2)$ : the Link's pace is the clock twice, and so is light's slowing.

**Derivation S.3** (the guard's edge and the horizon, Section 3.4). Inputs: Eq. (4). Steps: the lowest rotation of the band is at  $k_a = \pi$  on every axis, where  $2 \cos \omega = 2 - 2g(p_0/\Gamma)^2 - \frac{4 \text{num}}{3 \text{den}} \sum_a (p_a/\Gamma)^2$ ; the recurrence  $a_{\text{next}} + a_{\text{before}} = 2 \cos \omega a_{\text{now}}$  has no exponentially growing solution iff  $|2 \cos \omega| \leq 2$ ; at the edges  $\pm 2$  a linearly growing solution exists ( $u_t = \pm t$  from  $u_0 = 0, u_1 = 1$ ), which the guard admits, light's checkerboard mode sitting at  $-2$ ; for a negative numerator the lowest rotation is at  $k = 0$  and the same bound holds with  $|\text{num}|$ . With  $p_0 = p_a = p$  the condition  $2 \cos \omega \geq -2$  reads  $(p/\Gamma)^2 (2g + 4 \text{num}/\text{den}) \leq 4$ , that is  $(p/\Gamma)^2 \leq 2 \text{den}/(\text{den} + \text{num})$ , so the edge is  $P^2 = 2 \text{den} \Gamma^2 \text{div}(\text{den} + \text{num})$ , which is  $\Gamma^2$  for light. No horizon: under the composed paces  $p_0 = \Gamma(1 - 1/\Gamma)^c > 0$  at every content and  $p_a = p_0^2/\Gamma > 0$ , so  $R_a = 2 \text{num} p_a^2$  never vanishes and no Link closes; the quadratic truncation  $p_a = \Gamma(1 - 2x)$  vanished at  $x = 1/2$ , a horizon of the truncation. The frozen clock: at a content  $c$  the clock runs at  $N = (1 - 1/\Gamma)^c$  of the vacuum's,  $e^{-c/\Gamma}$  to the unit, so a deep well freezes a clock to the unit and its Node's share is not a number there.

**Derivation S.4** (the cube's group of 48 and its action, Section 3.1). Inputs: Rule3's line, whose six arrivals enter through the six Port coefficients  $R_{ij}$ , the two of an axis one  $R_a = 2 \text{num} p_a^2$  with no coefficient naming an axis beyond its own pace; the readings  $D, F$  and  $W$  of Eq. (1). The group: a map of the six Ports onto themselves that keeps opposite Ports opposite is a signed permutation of the three axes,  $\mathbf{x} \mapsto \mathbf{P}\mathbf{x}$  with  $\mathbf{P}$  a permutation matrix with signs; there are  $3! \cdot 2^3 = 48$  of them, the group  $O_h$  of the cube, of which the 24 with  $\det \mathbf{P} = +1$  are the rotations and the other 24 reflections and rotatory reflections; with the translations of the lattice they form the space group of the simple cubic lattice. Covariance: let  $\sigma$  be such a map and let the turned world carry the levels  $a_{\sigma(i)}^\sigma = a_i$  and the paces  $p_{\sigma(a)}^\sigma(\sigma(i)) = p_a(i)$ . The arrivals of the axis  $\sigma(a)$  at  $\sigma(i)$  in the turned world are the arrivals of the axis  $a$  at  $i$  in the original, in either order, and  $R_a$  is even in the axis's sign; so the sum  $\sum_a R_a(\text{arr}_{+a} + \text{arr}_{-a})$  is the same number at  $\sigma(i)$  as at  $i$ ,  $S$  and  $w$  are the same, and the division gives the same quotient and remainder: the step commutes with  $\sigma$ , and a world turned by  $\sigma$  runs as the turned world. A translation commutes for the same reason. The readings' kinds:  $D_i$  is one number at a Node with no axis in it, carried to  $\sigma(i)$  unchanged, a scalar. The currents  $F_{ij}$  are attached to the Links and carried with them, and the momentum density  $P_a(i) = (F_{i,i+a} - F_{i,i-a})/\text{num}$  changes sign where  $\sigma$  reverses the axis  $a$  and is carried to the image axis otherwise, a vector;  $W_i$  carries no axis and is carried unchanged, and it changes sign under the reflection of a plane record's sense,  $\text{im} \mapsto -\text{im}$ , a pseudoscalar in that sense. The click  $(R, t)$  names a region and an interval and is carried to the turned region, a scalar. The time reversal: exchanging now and before leaves  $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i$  unchanged once next and before exchange their roles, since  $D$  is symmetric in the pair (next, before); it reverses  $F_{ij}$  and  $W_i$ , which are antisymmetric in (now, before); and it carries the step to its inverse by Theorem 1. The rational rotation: the two parts of a plane record obey the same linear line with the same coefficients, so any linear combination of them does; the combination  $(\text{re}', \text{im}') = ((p\text{re} - q\text{im})/r, (q\text{re} + p\text{im})/r)$  with  $p^2 + q^2 = r^2$  carries solutions to solutions, and its Wronskian is  $(p^2 + q^2)/r^2$  times the old one, unchanged. What is not in the group: a boost changes the record's wave number and rotation together, which no permutation of Ports does, and a rotation by an angle other than the cube's carries a Node off the lattice; so neither is a symmetry, and Lorentz's factor, where the paper finds it, is a property of the clicks (Derivation S.21).

## Exact properties (Section 4)

**Derivation S.5** (the share's change is the currents, Eq. (10)). Inputs: the share at the vacuum's paces,  $e_i = 3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num} \text{now}_i S_6(\text{before})_i$ , with  $S_6$  the six-neighbour sum; Rule3 at the vacuum's paces, where  $S = 0$ ,  $R_a = 2 \text{num} \Gamma^2$  and  $w = 6 \text{den} \Gamma^2$ , so that  $3 \text{den}(\text{next}_i + \text{before}_i) = \text{num} S_6(\text{now})_i + \rho_i$  with  $\rho_i = (r_i - r'_i)/(2\Gamma^2)$  the

remainders' term (from  $6 \text{ den } \Gamma^2(\text{next} + \text{before}) = 2 \text{ num } \Gamma^2 S_6(\text{now}) + r - r'$ ). Steps: after the step the pair (now, before) becomes (next, now), so

$$e'_i - e_i = 3 \text{ den } (\text{next}_i^2 - \text{before}_i^2) - \text{num} [\text{next}_i S_6(\text{now})_i - \text{now}_i S_6(\text{before})_i].$$

With  $3 \text{ den } (\text{next}_i + \text{before}_i) = \text{num } S_6(\text{now})_i + \rho_i$  the first term is  $(\text{next}_i - \text{before}_i)(\text{num } S_6(\text{now})_i + \rho_i)$ , and the bracket cancels its  $\text{next}_i$  part:

$$\begin{aligned} e'_i - e_i &= \text{num} [\text{now}_i S_6(\text{before})_i - \text{before}_i S_6(\text{now})_i] + (\text{next}_i - \text{before}_i) \rho_i \\ &= \sum_{j \sim i} \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j) + (\text{next}_i - \text{before}_i) \rho_i, \end{aligned}$$

the six currents  $F_{ij}$  into the Node plus the remainder term. At any paces: with the weighted share  $e_i = [w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_j R_{ij} \text{before}_j] / (2p_i^2)$  and the line without its division,  $w \text{next}_i^* = \sum_j R_{ij} \text{now}_j + S_i \text{now}_i - w \text{before}_i$ , the difference  $e_i(\text{next}_i^*, \text{now}) - e_i(\text{now}, \text{before})$  is  $\sum_j (q_{ij}^2 / \Gamma^2) \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j)$  with no anisotropy term, the  $S_i$  terms and the Node's terms cancelling as in Theorem 3's proof, since  $R_{ij} / (2p_i^2) = \text{num } q_{ij}^2 / \Gamma^2$  is the same from both ends. The integer step's  $\text{next} = \text{next}_i^* + (r - r') / w$  adds  $-(\text{next}_i - \text{before}_i)(r'_i - r_i) / (2p_i^2)$ , because  $w(\text{next}_i + \text{next}_i^*) - S_i \text{now}_i - \sum_j R_{ij} \text{now}_j = w(\text{next}_i - \text{before}_i)$ . Check: exact fractions on a periodic chain of five Nodes at  $[2, 3]$ ,  $\Gamma = 10$ , with contents 0 to 3, Link factors 5 to 10 and random levels: the identity holds at every Node, with and without the division; and the automated reviewer's witness, a uniform  $\text{now} = 3$  over  $\text{before} = 0$  at  $\Gamma = 10$ ,  $q = 5$ , reads 81 per Node where the mean-of-paces form read 324 (an independent automated check).

**Derivation S.6** (the total share is non-negative, Section 4.4). Inputs:  $e_i$  as above on a closed or periodic GameBoard;  $S_6$  symmetric with eigenvalues  $2 \sum_a \cos k_a \in [-6, 6]$ . Steps:  $\sum_i e_i = 3 \text{ den } (|\mathbf{n}|^2 + |\mathbf{b}|^2) - \text{num } \mathbf{n} \cdot S_6 \mathbf{b}$ , and  $|\mathbf{n} \cdot S_6 \mathbf{b}| \leq \|S_6\| |\mathbf{n}| |\mathbf{b}| \leq 6 |\mathbf{n}| |\mathbf{b}| \leq 3(|\mathbf{n}|^2 + |\mathbf{b}|^2)$ , so  $\sum_i e_i \geq 3(\text{den} - |\text{num}|)(|\mathbf{n}|^2 + |\mathbf{b}|^2) \geq 0$  for  $|\text{num}| \leq \text{den}$ . Equality needs  $|\text{num}| = \text{den}$  and  $\mathbf{b}$  an eigenvector of  $S_6$  at  $\pm 6$  with  $\mathbf{n} = \pm \mathbf{b}$ : the static uniform level of light, or on an even periodic board its checkerboard,  $\text{now} = (-1)^{x+y+z}$  with  $\text{before} = -\text{now}$ , whose shares are 0. A Node's share: at fixed  $\text{before}_i$  and  $S_6(\text{before})_i = s$ ,  $e_i$  is a quadratic in  $\text{now}_i$  with the minimum  $3 \text{ den } \text{before}_i^2 - \text{num}^2 s^2 / (12 \text{ den})$ , negative iff  $|s| > 6(\text{den}/\text{num}) |\text{before}_i|$ ; a plane record has  $|s|/|\text{before}_i| = 2 \sum_a \cos k_a \leq 6$  and is never negative, a lay's edge can be. A region: pairing each boundary Link's cross term and bounding  $2|\text{now}_i \text{before}_j| \leq \text{now}_i^2 + \text{before}_j^2$ ,  $\sum_{i \in R} e_i \geq 3(\text{den} - \text{num}) \sum_R (\text{now}_i^2 + \text{before}_i^2) - \frac{\text{num}}{2} \sum_{\text{boundary Ports}} (\text{now}_j^2 + \text{before}_j^2)$  over the outside neighbours  $j$ : a region's share is negative at most by half a level-square per boundary Port in the current's units.

**Derivation S.7** (the tension of a plane wave, Section 4.3). Inputs: the stress reading  $G_{aa}(i) = \text{now}_i(\text{now}_{i+2a} - \text{now}_i) - \text{now}_{i+a}(\text{now}_{i+a} - \text{now}_{i-a})$ . Steps: for  $\text{now}_i = b \cos(kx_i + \phi)$  along the axis, with  $c_n = \cos(kx_i + nk + \phi)$ ,  $G = b^2[c_0 c_2 - c_0^2 - c_1^2 + c_1 c_{-1}]$ . The identities  $c_0 c_2 = \frac{1}{2}[\cos(2kx + 2k + 2\phi) + \cos 2k]$ ,  $c_1 c_{-1} = \frac{1}{2}[\cos(2kx + 2\phi) + \cos 2k]$ ,  $c_0^2 = \frac{1}{2}[1 + \cos(2kx + 2\phi)]$  and  $c_1^2 = \frac{1}{2}[1 + \cos(2kx + 2k + 2\phi)]$  cancel the phase-dependent parts, leaving  $G = b^2(\cos 2k - 1) = -2b^2 \sin^2 k$  at every Node and interval. So  $T_{aa} = -\text{num} (G(i) + G(i - a)) \text{div } 2 = 2 \text{ num } b^2 \sin^2 k$ , and for a wave with the components  $k_a, k_b$  the cross reading is  $G_{ab} = -2b^2 \sin k_a \sin k_b$ .

## Measurement (Section 5)

**Derivation S.8** (the region's reading factor and the draw's scatter, Section 5.3). Steps: a region of  $n$  consecutive rows sums a fringe  $\cos(2\pi y / \Lambda)$  to  $\text{Re} \sum_{y=0}^{n-1} e^{2\pi i y / \Lambda} = \text{Re} \frac{1 - e^{2\pi i n / \Lambda}}{1 - e^{2\pi i / \Lambda}}$ , whose modulus is  $\sin(\pi n / \Lambda) / \sin(\pi / \Lambda)$ ; over  $n$  rows the fringe's modulation is kept with the factor  $\sin(\pi n / \Lambda) / (n \sin(\pi / \Lambda))$ , 0.90 at  $n = 4$ ,  $\Lambda = 16$ , and 0 at  $\Lambda = n$ . A frequency read as a click rate over  $N$  intervals resolves  $\Delta\omega \geq 2\pi / N$ , the window's own resolution.  $N$  quanta drawn independently by the shares  $p_i$  give each region a binomial count,  $N p_i \pm \sqrt{N p_i (1 - p_i)}$ .

**Derivation S.9** (the Wronskian's conservation and its current, Section 5.5). Inputs: the two parts of a plane record step by the same symmetric read matrix  $\mathbf{M}$  (the paper's Theorem 3),  $\mathbf{re}_{\text{next}} = \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$  and the same for  $\mathbf{im}$ , with the weights  $\mathbf{D}$  of the conserved form,  $\mathbf{D} \mathbf{M}$  symmetric. Steps: let  $W = \mathbf{re}_{\text{now}}^\top \mathbf{D} \mathbf{im}_{\text{before}} - \mathbf{im}_{\text{now}}^\top \mathbf{D} \mathbf{re}_{\text{before}}$ . After the step it is  $\mathbf{re}_{\text{next}}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{next}}^\top \mathbf{D} \mathbf{re}_{\text{now}} = (\mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}})^\top \mathbf{D} \mathbf{im}_{\text{now}} - (\mathbf{M} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{before}})^\top \mathbf{D} \mathbf{re}_{\text{now}}$ ; the  $\mathbf{M}$  terms are  $\mathbf{re}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{im}_{\text{now}} - \mathbf{im}_{\text{now}}^\top \mathbf{M}^\top \mathbf{D} \mathbf{re}_{\text{now}}$ , which vanish since  $\mathbf{M}^\top \mathbf{D} = \mathbf{D} \mathbf{M}$  is symmetric, and the rest is  $W$  again. Per Node the same cancellation leaves the Link terms: the weighted Wronskian  $W_i / p_i^2$  changes by  $(2 \text{ num} / w) (q_{ij}^2 / \Gamma^2) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  summed over the Node's six Ports at the levels now, the Wronskian's current, which for the integer  $W_i$  at the vacuum's paces is  $(\text{num} / 3 \text{ den}) (\text{im}_i \text{re}_j - \text{re}_i \text{im}_j)$  per Port. Check: a periodic chain of three Nodes at  $\Gamma = 2$ ,  $[1, 1]$ ,  $\mathbf{re}_{\text{now}} = (0, 3, 0)$ ,  $\mathbf{im}_{\text{now}} = (3, 0, 0)$  and zeros before steps to  $(1, 4, 1)$  and  $(4, 1, 1)$  with no remainder;  $W_0$  goes from 0 to 3,  $W_0 / p_0^2$  by  $3/4$ , and the current is  $(2/24) \cdot 9 = 3/4$  (an independent automated check's witness); exact

fractions on a chain of five Nodes with contents and Link factors confirm the identity at every Node. For the circular record  $\text{re} = A \cos \omega t$ ,  $\text{im} = e A \sin \omega t$  at one Node,  $W = A \cos \omega t \cdot e A \sin \omega(t-1) - e A \sin \omega t \cdot A \cos \omega(t-1) = -e A^2 \sin \omega$ , constant. Domain: the identity is the line's,  $\mathbf{re}_{\text{next}} = \mathbf{M} \mathbf{re}_{\text{now}} - \mathbf{re}_{\text{before}}$  exactly. The integer step's division adds  $\epsilon = (r - r')/w$  to each part, and the remainder term  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  is not zero in general: at the vacuum's pace  $S = 0$  and  $w a_{\text{next}} = 12 \text{num} \Gamma^2 a_{\text{now}} - w a_{\text{before}}$  for a uniform level, so at  $[2, 3]$  the parts  $(1, 3)$  with zeros before step to  $([4/3], [4]) = (1, 4)$  and  $W$  goes from 0 to  $1 \cdot 3 - 4 \cdot 1 = -1$ . An exactly conserved integer charge is open.

**Derivation S.10** (Bell's correlation under the declared credit, Section 5.7 and Section 9.4). Inputs: two parts with the products  $M_1, M_2$  of their amplitudes at the two sides' ports; the port coefficients  $e(+) = (\cos a, \sin a)$ ,  $e(-) = (-\sin a, \cos a)$  on side A and the same with  $b$  on side B; the joint share  $J(p, q) = (\sum_k e_k(p) e_k(q) M_k)^2$ ;  $E = \sum_{p,q} s(p)s(q) J(p, q) / \sum_{p,q} J(p, q)$  with  $s(\pm) = \pm 1$ . Steps: write  $c_a = \cos a$ ,  $s_a = \sin a$  and the same for  $b$ . The four sums are  $M_1 c_a c_b + M_2 s_a s_b$ ,  $M_1 s_a s_b + M_2 c_a c_b$ ,  $-M_1 c_a s_b + M_2 s_a c_b$  and  $-M_1 s_a c_b + M_2 c_a s_b$  for  $(+, +)$ ,  $(-, -)$ ,  $(+, -)$  and  $(-, +)$ . Their squares sum to  $(M_1^2 + M_2^2)(c_a^2 c_b^2 + s_a^2 s_b^2 + c_a^2 s_b^2 + s_a^2 c_b^2) = M_1^2 + M_2^2$ , the cross terms cancelling in pairs. The signed sum is  $(M_1^2 + M_2^2)(c_a^2 - s_a^2)(c_b^2 - s_b^2) + 8 M_1 M_2 c_a s_a c_b s_b$ , so

$$E(a, b) = \cos 2a \cos 2b + \rho \sin 2a \sin 2b, \quad \rho = \frac{2 M_1 M_2}{M_1^2 + M_2^2},$$

and the marginal  $P(A+) = [(M_1 c_a c_b + M_2 s_a s_b)^2 + (-M_1 c_a s_b + M_2 s_a c_b)^2] / (M_1^2 + M_2^2) = (M_1^2 c_a^2 + M_2^2 s_a^2) / (M_1^2 + M_2^2)$ , independent of  $b$ . For equal parts  $\rho = 1$  and  $E = \cos 2(a - b)$ . The integer settings: with  $e(+) = (p, q)$  and  $e(-) = (-q, p)$  unnormalised and equal parts,  $J(+, +) = J(-, -) = (p_{APB} + q_{AQB})^2$  and  $J(+, -) = J(-, +) = (p_{AQB} - q_{APB})^2$ , whose sum is  $2(p_A^2 + q_A^2)(p_B^2 + q_B^2)$  by Lagrange's identity, so  $E = [(p_{APB} + q_{AQB})^2 - (p_{AQB} - q_{APB})^2] / [(p_A^2 + q_A^2)(p_B^2 + q_B^2)]$ , a rational number. At  $a = (1, 0)$ ,  $a' = (1, 1)$ ,  $b = (12, 5)$ ,  $b' = (5, 12)$ :  $E(a, b) = (144 - 25)/169 = 119/169$ ,  $E(a, b') = (25 - 144)/169 = -119/169$ ,  $E(a', b) = (17^2 - 7^2)/338 = 120/169$ ,  $E(a', b') = 120/169$ , and  $S = E(a, b) - E(a, b') + E(a', b) + E(a', b') = 478/169 = 2.8284$ . At the ideal angles  $S = \sqrt{2}(1 + \rho)$ ; at the integer settings  $S = (238 + 240\rho)/169$ . The two local credits: with the pair's phase  $\theta$  uniform, a credit of each side by the shares of its ports gives  $E = \langle \cos 2(a - \theta) \cos 2(b - \theta) \rangle_\theta = \frac{1}{2} \cos 2(a - b)$  and  $S = \sqrt{2}$ ; a credit by the sign of the larger port gives  $E = \langle \text{sgn} \cos 2(a - \theta) \text{sgn} \cos 2(b - \theta) \rangle_\theta = 1 - 4|a - b|/\pi$ , the sawtooth, whose sum at the four settings is exactly 2. Check: the averages over  $\theta$  on a grid of  $2 \times 10^5$  points give  $S = 1.4142$  and  $2.0000$ , and the contrast credit, which discards the pairs near the contrast's zero,  $S = 2.8284$  at the efficiency  $2/\pi$ .

## The families (Section 6)

**Derivation S.11** (the reach of a field with a gap, Section 6.4). Inputs: a field's rest outside its source, the line with  $a_{\text{next}} = a_{\text{before}} = a_{\text{now}} = a$ :  $6 \text{den } a = \text{num } S_6(a)$  at the vacuum's pace, that is  $\text{num}(S_6 - 6)a = 6(\text{den} - \text{num})a$ . Steps: for a rest falling along one axis as  $e^{-\kappa r}$  and uniform across,  $S_6(a) = (2 \cosh \kappa + 4)a$ , so  $6 \text{den} = \text{num}(2 \cosh \kappa + 4)$  and  $\cosh \kappa = 3 \text{den}/\text{num} - 2$ . For  $\text{num}$  just under  $\text{den}$ ,  $\cosh \kappa = 1 + \kappa^2/2 + \dots$ , so  $\kappa^2 = 6 \text{den}/\text{num} - 6 = 6(\text{den} - \text{num})/\text{num}$  and the range is  $R = 1/\kappa = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$ . Links: 20.00 at  $[2400, 2401]$ ;  $[1, 2]$  has  $\cosh \kappa = 4$ ,  $e^{-\kappa} = 0.127$ . The gap is also the rest rotation of the field's own travelling events,  $\cos \omega_g = \text{num}/\text{den}$ ,  $\omega_g^2 = 2(\text{den} - \text{num})/\text{den} + \dots$ , whose Compton length at light's speed,  $c/\omega_g = \sqrt{\text{den}/(6(\text{den} - \text{num}))} = R\sqrt{\text{den}/\text{num}}$ , is the range to the gap's first order: Yukawa's identity with no input.

**Derivation S.12** (the two speeds, Section 6.5). Steps: light at a content  $x$  travels at  $(1 - 2x)/\sqrt{3}$  (Derivation S.2). Under the law's line the massless field reads the content as every family does, its own level among it, so its travelling events have the same paces and the two arrive together at every content, the difference of their Shapiro delays 0. With the massless field reading no content, as an earlier version of the implementation had it, its events travel at  $1/\sqrt{3}$  and over  $L$  Links the two arrive  $L\sqrt{3}[1/(1 - 2x) - 1] = 2xL\sqrt{3} + O(x^2)$  intervals apart at long wavelength, against GW170817's bound on  $\gamma_{\text{GW}} - \gamma_{\text{EM}}$ , between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$ ; under the line the measurement bounds no vacuum content.

## Bodies (Section 7)

**Derivation S.13** (the bound body: the fixed point and the functional, Section 7.2). Inputs: Rule3's line at the paces of the wells the record sources; a standing record  $a_{i,t} = \phi_i \cos(\omega_b t)$  with one rotation at every Node; the clock's pace  $(p_0/\Gamma)^2 = 1 - 2\Phi_i/\Gamma + \dots$  at a well  $\Phi_i$  (in levels). The fixed point's equation: with the remainders dropped the line gives at every Node

$$2 \cos \omega_b \phi_i = \frac{1}{w} \left[ \sum_a R_a(i) (\phi_{i+a} + \phi_{i-a}) + S_i \phi_i \right] = (\mathbf{M}\phi)_i,$$

so  $\phi$  is an eigenvector of the read matrix  $\mathbf{M}$  and the standing record is its top mode, the maximiser of  $\phi^\top \mathbf{D} \mathbf{M} \phi$  on the weighted sphere  $\phi^\top \mathbf{D} \phi = 1$ ,  $\mathbf{D} \mathbf{M}$  symmetric under the law's read coefficient (S.5) (the generator of Section 7.1 iterates this read act). To the first order in  $\Phi/\Gamma$ ,  $S_i/w = 2 - 2g(1 - 2\Phi_i/\Gamma) - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2 = 2 \cos \omega_0 + 4g\Phi_i/\Gamma - \frac{2\text{num}}{3\text{den}} \sum_a (p_a/\Gamma)^2$  and  $R_a/w = \frac{\text{num}}{3\text{den}} (p_a/\Gamma)^2$ , so, keeping the band's term at the vacuum's Link pace (its product with the well is of the second order),

$$(\mathbf{M}\phi)_i = 2 \cos \omega_0 \phi_i + \frac{4g}{\Gamma} \Phi_i \phi_i - \frac{\text{num}}{3\text{den}} (-\Delta^2 \phi)_i, \quad (-\Delta^2 \phi)_i = 6\phi_i - \sum_a (\phi_{i+a} + \phi_{i-a}).$$

The functional: the well is sourced by the record itself,  $\Phi_i = \sum_r \sigma_r (M/E_r) \sum_j G_{r,ij} \phi_j^2$  for a body of  $M$  quanta with the counts  $c_j = M\phi_j^2$ . Let

$$F[\phi] = \frac{\text{num}}{3\text{den}} \sum_{\text{Links}} (\phi_i - \phi_j)^2 - \frac{2g}{\Gamma} \sum_r \sigma_r \frac{M}{E_r} \sum_{ij} \phi_i^2 G_{r,ij} \phi_j^2,$$

the paper's Eq. (12). Its gradient is  $\partial F/\partial \phi_i = \frac{2\text{num}}{3\text{den}} (-\Delta^2 \phi)_i - \frac{8g}{\Gamma} \Phi_i \phi_i$ , using the symmetry of  $G_r$  (each well counted once, Hartree's form, is what makes the factor  $2g$  in  $F$  give  $4g$  in the gradient). So  $\partial F/\partial \phi = 2\lambda\phi$  on the sphere is exactly the fixed point's equation with  $\lambda = 2 \cos \omega_0 - 2 \cos \omega_b$ : a body is a critical point of  $F$  on the sphere, and the virial and the second variation along the dilation are Theorem 4 of the paper. The scaling powers: under  $\phi_R(\mathbf{x}) = R^{-d/2} \phi(\mathbf{x}/R)$  the sum over Links of  $(\phi_i - \phi_j)^2$  is the lattice form of  $\int |\nabla \phi|^2$  and scales as  $R^{-2}$ , the pressure  $K/R^2$ . The well energy  $\sum_{ij} \phi_i^2 G_{ij} \phi_j^2$  is  $\iint \rho(\mathbf{x}) G(\mathbf{x} - \mathbf{y}) \rho(\mathbf{y})$  with  $\rho = \phi^2$  of unit mass, which scales as  $R^{-(d-2)}$  for the massless kernel  $G \sim 1/r$  and for a screened kernel within its reach, and as  $R^{-d}$  for a screened kernel beyond its reach, where  $G$  acts as a point kernel and the energy is  $\int \rho^2$ : the powers  $s_r$  of Theorem 4. The cloud's size: for the massless kernel in three dimensions  $F(R) = a/R^2 - c/R$  with  $a = \frac{\text{num}}{3\text{den}} \int |\nabla \phi|^2$  and  $c = \frac{2g}{\Gamma} \frac{M}{E} \iint \frac{\rho \rho'}{4\pi |\mathbf{x} - \mathbf{y}|} \cdot 3$  at the unit size, so the minimum is at  $R = 2a/c \propto \Gamma E/(gM)$ : the radius falls as  $1/M$  and grows with  $\Gamma$ , the paper's  $5.25 \Gamma E_g/C$  up to the shape's constants, which the generator computes.

**Derivation S.14** (the invariant and the count's drift, Eq. (13)). Inputs: the one-Node line  $a_{t+1} + a_{t-1} = 2 \cos \omega_t a_t$  with  $\omega_t$  changing slowly. Steps: at fixed  $\omega$  the form  $D_t = a_t^2 - a_{t+1} a_{t-1} = a_t^2 + a_{t-1}^2 - 2 \cos \omega a_t a_{t-1}$  is conserved and equals  $A^2 \sin^2 \omega$  for  $a_t = A \cos(\omega t + \phi)$ ; in the plane  $(a_t, a_{t-1})$  the step is a linear map of determinant 1 whose orbits are the ellipses  $D = \text{const}$ , of area  $\pi D / \sin \omega$ . A slow change of  $\omega$  keeps the area enclosed by the orbit, the action of the map, so  $D / \sin \omega = A^2 \sin \omega$  is the adiabatic invariant: the discrete form of  $E/\omega$ . Equivalently, the ansatz  $a_t = C \cos(\sum_{s \leq t} \omega_s + \phi) / \sqrt{\sin \omega_t}$  satisfies the line to the first order in  $\omega_{t+1} - \omega_t$ . Check: the recurrence stepped 20,000 intervals with  $\omega$  lowered from 0.841 to 0.600 keeps  $D / \sin \omega$  within  $4 \times 10^{-5}$  of its value while  $D$  falls by a quarter. The drift: the share at a Node in quanta is  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b / T$  (Eq. (9) at the Node's paces), so with  $A^2 \sin \omega_b$  fixed,  $N(t_2)/N(t_1) = [\sin \omega_b(t_2) / \sin \omega_b(t_1)] [p_i^2(t_1)/p_i^2(t_2)]$ , Eq. (13). Planck's relation: the share  $A^2 \sin^2 \omega / T$  is the invariant  $I = A^2 \sin \omega / T$  times  $\sin \omega$ . The body's click: under the declaration that one click is one quantum's change in the absorbing body (Section 7.4), the absorber answers at its own rotation by the two-mode line of S.16, one transition per click from the mode  $\omega_i$  to the mode  $\omega_j$ , and the share it takes is  $T(\sin \omega_j - \sin \omega_i) = T \sin(\omega_j - \omega_i) [1 - \omega_i \omega_j / 2 + O(\omega^4)]$ , by  $\sin \omega_j - \sin \omega_i = 2 \cos \frac{\omega_i + \omega_j}{2} \sin \frac{\omega_j - \omega_i}{2}$ : one photon's share  $T \sin \omega_L$  at the line  $\omega_L = \omega_j - \omega_i$ , to a relative  $\omega_0^2/2 < 10^{-18}$  at nature's gap (S.27). So a body counts photons,  $I$  per click, and reads  $\hbar \omega$  per click with  $T$  for  $\hbar$ ; the share is the energy a click carries, and through a deepening well the number of quanta,  $I$ , is kept.

**Derivation S.15** (the clock's click rate, Section 7.4). Inputs: Eq. (4) at  $k = 0$  with  $p_0^2 = \Gamma^2(1 - 2U + 2U^2)$ :  $\cos \omega(U) = 1 - g(1 - 2U + 2U^2) = \cos \omega_0 + 2gU - 2gU^2$ . Steps: to first order  $d(\cos \omega) = 2g dU = -\sin \omega_0 d\omega$ , so  $d\omega/\omega_0 = -[2g/(\omega_0 \sin \omega_0)] dU = -f(\omega_0) dU$  with  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ ;  $f = 1.063$  at  $[2, 3]$ , and as  $\omega_0 \rightarrow 0$ ,  $1 - \cos \omega_0 \rightarrow \omega_0^2/2$  and  $f \rightarrow 1$ , the rate  $1 - U$  of the clock's pace alone. The light clock: a packet between two declared faces  $L$  Links apart at the content  $x$  has the round trip  $2L/v_g = 2L\sqrt{3}/N^2$  (Derivation S.2), so its rate falls by  $2x$  to the first order; between two bodies, which stand  $L/h$  Links apart under the principle, the round trip is  $2L\sqrt{3}/N$  (Derivation S.31).

**Derivation S.16** (the resonance at twice the rotation, Section 7.4). Inputs: the one-Node line with the rotation modulated by a level  $b_L \cos \omega_L t$  read into the content, the law's two-mode line; light's declared read is the holder's rotation (Section 5.5), under which S.43 derives the transfer line:  $2 \cos \omega(t) = 2 - 2g(1 - 2x + 2x^2)$  with  $x = x_0 + b_L \cos(\omega_L t)/\Gamma$ , so  $d(2 \cos \omega)/dx = 4g(1 - 2x)$  and  $a_{\text{next}} + a_{\text{before}} = [2 \cos \omega_b + \epsilon \cos(\omega_L t + \vartheta)] a_{\text{now}}$  with  $\epsilon = 4g(1 - 2x_0) b_L/\Gamma$ . Steps: write  $a_t = \text{Re}[z_t e^{i\omega_b t}]$  with  $z$  slowly varying,  $z_{t \pm 1} = z_t \pm z'$ . Then  $a_{t+1} + a_{t-1} = \text{Re}[e^{i\omega_b t} (2 \cos \omega_b z_t + 2i \sin \omega_b z')]$ , and the product  $\epsilon \cos(\omega_L t + \vartheta) \text{Re}[z_t e^{i\omega_b t}]$  has, at  $\omega_L = 2\omega_b$ , the resonant part  $(\epsilon/2) \text{Re}[\bar{z}_t e^{i\vartheta} e^{i\omega_b t}]$ . Equating the slowly varying coefficients of  $e^{i\omega_b t}$ ,  $2i \sin \omega_b z' = (\epsilon/2) e^{i\vartheta} \bar{z}$ , so  $z' = -i\epsilon e^{i\vartheta} \bar{z}/(4 \sin \omega_b)$ ; this antilinear map  $z \mapsto \lambda \bar{z}$  with  $|\lambda| = \mu$  has the real eigenvalues  $\pm \mu$  on two lines of the  $z$  plane, so the amplitude grows as  $e^{\mu t}$  along one line and the share,

quadratic in the amplitude, as  $e^{2\mu t}$ . Detuned by  $\Delta = \omega_L - 2\omega_b$ , the substitution  $z = \zeta e^{i\Delta t/2}$  gives  $\zeta' = -i(\Delta/2)\zeta - i\mu e^{i\vartheta}\bar{\zeta}$ , whose exponents are  $\pm\sqrt{\mu^2 - \Delta^2/4}$ : growth only for  $|\Delta| < 2\mu$ , a full width  $\epsilon/\sin\omega_b$ . Check: the recurrence stepped 20,000 intervals at  $2\cos\omega_b = 1.4317$  and  $\epsilon = 0.013$  grows at 0.00461 per interval against the formula's first-order 0.004654, the difference of the order of  $\epsilon$ . The two-mode line: with two bound modes the same modulation multiplies  $a_t = \text{Re}[z_i e^{i\omega_i t} + z_j e^{i\omega_j t}]$ , and its product with  $\cos\omega_L t$  carries the frequencies  $\omega_i \pm \omega_L$ . The term resonant with the mode  $j$  from the mode  $i$  is at  $\omega_L = \omega_j - \omega_i$ , a transfer between the two modes, and the term that drives both from the conjugate of the other is at  $\omega_L = \omega_i + \omega_j$ , of which the one-mode case above is the degenerate case  $i = j$  (at  $i = j$  the difference line is  $\omega_L = 0$ , a static shift and no line). The transfer's rate,  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin\omega_i \sin\omega_j})$ , follows by the same slowly varying coefficients with the two modes'  $2i\sin\omega_i z'_i$  and  $2i\sin\omega_j z'_j$  coupled through  $\epsilon_{ij}$ , a Rabi oscillation; under light's declared read, the rotation, S.43 derives the transfer line.

**Derivation S.17** (the cross current of two standing records, Section 7.5). Inputs: two standing records  $a_t = A\cos(\omega_1 t + \alpha)$  and  $b_t = B\cos(\omega_2 t + \beta)$  at the two ends of a Link. Steps: the current  $\text{num}(a_t b_{t-1} - a_{t-1} b_t)$  is  $\text{num}AB[\cos(\omega_1 t + \alpha)\cos(\omega_2 t - \omega_2 + \beta) - \cos(\omega_1 t - \omega_1 + \alpha)\cos(\omega_2 t + \beta)]$ , a sum of cosines of  $(\omega_1 - \omega_2)t$  and  $(\omega_1 + \omega_2)t$  with the constant parts  $\frac{1}{2}\cos(\alpha - \beta + \omega_2)$  and  $\frac{1}{2}\cos(\alpha - \beta - \omega_1)$  only when  $\omega_1 = \omega_2$ ; for  $\omega_1 \neq \omega_2$  its mean over the beat is 0, so no quantum passes at the first order. The reach: two bodies' tails are  $a_1 e^{-\kappa d_1}$  and  $a_2 e^{-\kappa d_2}$  at a Link  $d = d_1 + d_2$  Links from the two centres, so the current there is of the order  $a_1 a_2 e^{-\kappa d}$  per interval and falls under one quantum,  $T$ , at  $d = \ln(a_1 a_2 / T) / \kappa$ .

## The forms of nature (Section 8)

**Derivation S.18** (the fall and the Newtonian potential the orbits measure, Eq. (14)). Inputs: Eq. (4) at a content  $U$  for a reader at the weight 1,  $\cos\omega(k, U) = 1 - g(1 - 2U + 2U^2) - \frac{\text{num}}{3\text{den}}(1 - 2U)^2(1 - \cos k)$  along the motion. Steps: at small  $k$  and  $U$ ,  $\omega(k, U) = \omega_0(U) + k^2/(2m^*)$  with the inertia  $m^* = 3\text{den}\sin\omega_0/\text{num} = 3\tan\omega_0$  (from  $\sin\omega_0 d\omega = \frac{\text{num}}{3\text{den}} k dk$ ), so the group velocity is  $v = k/m^*$ . The rays of  $\omega(k, x)$  give  $dk/dt = -\partial\omega/\partial x = -(d\omega_0/dU)\nabla U = (2g/\sin\omega_0)|\nabla U|$  toward the well (Derivation S.15), so the acceleration is

$$a = \frac{1}{m^*} \frac{dk}{dt} = \frac{\text{num}}{3\text{den}\sin\omega_0} \cdot \frac{2g}{\sin\omega_0} |\nabla U| = \frac{2\cos\omega_0(1 - \cos\omega_0)}{3\sin^2\omega_0} |\nabla U| = \frac{2\cos\omega_0}{3(1 + \cos\omega_0)} |\nabla U|,$$

which in the gap  $\epsilon = 1 - \cos\omega_0$  is  $[(1 - \epsilon)/(3(1 - \epsilon/2))]| \nabla U|$ : 0.2667 at  $[2, 3]$  and  $1/3$  as the gap closes. Nature's fall is  $a = c^2 |\nabla U_K|$  with  $U_K = GM_K/(c^2 r)$  and  $c^2 = 1/3$  here, so  $U_K = [2\cos\omega_0/(1 + \cos\omega_0)]U$ , Eq. (14), 0.800 at  $[2, 3]$ .

**Derivation S.19** (the post-Newtonian parameters against the Newtonian potential the orbits measure, Section 8). Inputs: Eq. (14),  $U = \lambda U_K$  with  $\lambda = (1 + \cos\omega_0)/(2\cos\omega_0) = (\text{den} + \text{num})/(2\text{num})$ ; the three forms in  $U$ : the bending  $4U_b$ , the periapsis  $6\pi U_1/(a(1 - e^2))$ , the redshift  $f(\omega_0)U$ . Steps: the bending is  $4\lambda U_{K,b} = 2(1 + \text{den}/\text{num})U_{K,b}$ , and the post-Newtonian bending is  $(1 + \gamma)2GM/(c^2 b) = (1 + \gamma)2U_{K,b}$ , so  $1 + \gamma_K = 1 + \text{den}/\text{num}$ ,  $\gamma_K = \text{den}/\text{num}$ . The periapsis is  $\lambda$  times Einstein's  $6\pi U_{K,1}/(a(1 - e^2))$ , and the post-Newtonian advance is  $(2 - \beta + 2\gamma)/3$  of Einstein's, so  $(2 - \beta_K + 2\text{den}/\text{num})/3 = (\text{den} + \text{num})/(2\text{num})$ , whence  $\beta_K = 2 + 2\text{den}/\text{num} - 3(\text{den} + \text{num})/(2\text{num}) = (\text{den} + \text{num})/(2\text{num})$ . The redshift is  $f(\omega_0)\lambda U_K$ , and  $f\lambda = \frac{2(1 - \cos\omega_0)}{\omega_0 \sin\omega_0} \cdot \frac{1 + \cos\omega_0}{2\cos\omega_0} = \frac{\sin^2\omega_0}{\omega_0 \sin\omega_0 \cos\omega_0} = \frac{\tan\omega_0}{\omega_0}$ , so  $1 + \alpha_K = \tan\omega_0/\omega_0$ . At  $[2, 3]$ : 1.5, 1.25 and 1.329; all three tend to 1 as  $\omega_0 \rightarrow 0$ , and  $\gamma_K - 1 < 2 \times 10^{-5}$  requires  $\text{den}/\text{num} - 1 < 2 \times 10^{-5}$ .

**Derivation S.20** (Shapiro's delay, Section 8 (c)). Inputs: light's index  $n = 1 + 2U$  with  $U = \mu/r$  (Derivation S.2). Steps: the delay along a straight path at the impact parameter  $b$  from  $-x_1$  to  $x_2$  is  $\int (n - 1) dl/c = 2\sqrt{3}\mu \int_{-x_1}^{x_2} dl/\sqrt{l^2 + b^2} = 2\sqrt{3}\mu [\text{arsinh}(x_2/b) + \text{arsinh}(x_1/b)]$ , which for  $x_1, x_2 \gg b$  is  $2\sqrt{3}\mu \ln(4x_1 x_2/b^2)$ : Shapiro's delay with  $\gamma = 1$  in  $U$ , and with  $\gamma_K$  against  $U_K$  by the same factor as the bending, since the bending is the transverse derivative of the same integral.

**Derivation S.21** (the moving clock to the second order, Section 8 (e)). Inputs: the exact band of a free record of the pair  $[\text{num}, \text{den}]$  along an axis, from Eq. (1) at the vacuum's pace (S.1):  $\cos\omega = c_0(2 + \cos k)$  with  $c_0 = \text{num}/(3\text{den})$ ,  $\cos\omega_0 = 3c_0$ ; the rotation at the moving centre,  $\Omega(k) = \omega(k) - k\omega'(k)$ , the phase rate less  $k$  times the group velocity  $v = \omega'(k)$ , which is what a clock carried by the packet reads. Steps: write  $s = \sin\omega_0$ ,  $c = \cos\omega_0$  and  $u = 1 - \cos k = k^2/2 - k^4/24 + \dots$ ; then  $\cos\omega = c - c_0 u$ , and with  $\omega = \omega_0 + \delta$ ,  $\cos\omega = c - s\delta - c\delta^2/2 + \dots$ , so  $s\delta + c\delta^2/2 = c_0 u$  and

$$\omega(k) = \omega_0 + \alpha k^2 - \beta k^4 + O(k^6), \quad \alpha = \frac{c_0}{2s}, \quad \beta = \frac{c_0}{24s} + \frac{c c_0^2}{8s^3}.$$

Hence  $v = \omega' = 2\alpha k - 4\beta k^3$ ,  $\Omega = \omega - k\omega' = \omega_0 - \alpha k^2 + 3\beta k^4$ , and inverting  $v(k)$ ,  $k^2 = v^2/(4\alpha^2) + \beta v^4/(4\alpha^5) + \dots$ , so

$$\frac{\Omega}{\omega_0} = 1 - \frac{v^2}{4\alpha\omega_0} - \frac{\beta v^4}{16\alpha^4\omega_0} + O(v^6) = 1 - \frac{v^2}{2c_m^2} - \frac{2\omega_0}{\sin 2\omega_0} \cdot \frac{v^4}{8c_m^4} + O(v^6), \quad c_m^2 = 2\alpha\omega_0 = \frac{c_0\omega_0}{s} = \frac{\omega_0}{3\tan\omega_0},$$

where the fourth-order coefficient is  $\beta/(16\alpha^4\omega_0)$  over Lorentz's  $1/(8c_m^4)$ , which reduces with  $3c_0 = c$  to  $\omega_0(s/c + c/s) = \omega_0/(\sin\omega_0 \cos\omega_0) = 2\omega_0/\sin 2\omega_0$ . So the tick of a free quantum is Lorentz's factor at the family's own light speed  $c_m$  to the second order in  $v$  exactly, and at the fourth order it carries the factor  $2\omega_0/\sin 2\omega_0$ , 1.69 at  $[2, 3]$  and 1 as the gap closes: the full factor is reached only in the limit of a small rest rotation, the same limit in which the post-Newtonian parameters reach 1 (S.19). Check: with the exact band at  $[2, 3]$  the residual  $(f - 1 + v^2/(2c_m^2))/v^4$  at  $v = 0.02$  is  $-3.37$  against 1.6926 times Lorentz's  $-1.988$ , that is  $-3.365$ . A body: a bound body's centre moves on the body's own band,  $\omega_b(k)$ , whose rest rotation and curvature the generator computes from the fixed point (S.13); the same steps on that band give the same form with the body's  $c_b^2 = \omega_b/m_b^*$ , and the body's band is a computed input, not derived in closed form here.

**Derivation S.22** (de Broglie's relation, Section 8 (h)). Steps: from Derivation S.18,  $v = k/m^*$  with  $m^* = 3 \tan \omega_0$  at the vacuum's content, so  $k = m^*v = \omega_b v/c_m^2$  with  $c_m^2 = \omega_0/m^* = \omega_0/(3 \tan \omega_0)$ , the family's own light speed; 0.2508 at  $[2, 3]$ .

## The two slits (Section 9)

**Derivation S.23** (the fringe's spacing and the near-field minima, Section 9.2). Inputs: the gaps centred at the rows 18 and 30,  $d = 12$ ; the screen  $L = 24$  Links from the wall;  $\lambda = 8$ . Steps: the far-field spacing is  $\lambda L/d = 16$  Links. In the near field the minima are where the two gaps' paths to the screen differ by half a wavelength,  $\sqrt{L^2 + (y - 18)^2} - \sqrt{L^2 + (y - 30)^2} = \pm 4$ , whose solutions are  $y = 15.28$  and  $y = 32.72$ : the rows 15 and 33. Check: the two roots computed from the equation above.

## Calibration (Section 10)

**Derivation S.24** (the Link's bound from light's dispersion, Section 10.2). Inputs: the group speed  $v_g = (1/\sqrt{3})(1 - k^2/12)$  (Derivation S.1); the bound on a quadratic dependence of light's speed on its energy, MAGIC's  $E_{QG,2}$  above about  $6 \times 10^{10}$  GeV, that is  $|\delta v/v| = (3/2)(E/E_{QG,2})^2 < 4.5 \times 10^{-16}$  at  $E = 1$  TeV in the standard quadratic form. Steps:  $k^2/12 < 4.5 \times 10^{-16}$  gives  $k < 7.3 \times 10^{-8}$  per Link at 1 TeV; with  $k = 2\pi \text{Link}/\lambda$  and  $\lambda = 1.24 \times 10^{-18}$  m at 1 TeV, the Link is below  $k\lambda/(2\pi) = 1.4 \times 10^{-26}$  m.

**Derivation S.25** (Newton's constant as a reading, Eq. (18)). Inputs: the massless field's rest far from a static source of  $s$  quanta per interval,  $3s/(4\pi r)$  over the divisor  $E_g$ ; the clock's share  $U = c/\Gamma$ . Steps:  $U = 3s/(4\pi r \Gamma E_g)$ , and nature's  $U = GM/(c^2 r)$  with  $M = s$  in quanta and  $c^2 = 1/3$  gives  $G/c^2 = 3/(4\pi \Gamma E_g)$ , that is  $G_{\text{clock}} = 1/(4\pi \Gamma E_g)$ ; Kepler's  $G$ , read from the orbits of matter, is  $[2 \cos \omega_0/(1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (14).

**Derivation S.26** (the fine-structure constant as a coefficient of the file, Section 10.1). Inputs: the holder of the sign under its rotation (Section 5.5) with its declared write weight  $k$ , the written level  $k$  times the plain one and the turn  $\theta = \text{level}/\Gamma$ ,  $k = 1$  in the implementation's files; a circular record of a family at the vacuum's paces,  $z = Ae^{i\omega_s t}$ , the quantum of a family one unit of the invariant,  $2A^2 \sin \omega_s = T$ , with the share  $T \sin \omega_s$  and the Wronskian  $A^2 \sin \omega_s = T/2$  (S.9); the holder's rest far from a static source of  $s$  per interval,  $3s/(4\pi r)$  over its weight  $E_s$  (Derivation S.25's line); de Broglie's relation and light's  $c = 1/\sqrt{3}$  (S.1, S.22). Steps: (a) One quantum writes  $s = W/(E_s T) = 1/(2E_s)$  per interval, for every family, since the sourcing family's denominator enters no write (the count is the share over the wall  $3 \text{den}_s T$  and the source the form over  $E_s T$ , the same number); so  $|q_p| = |q_e|$  with nothing declared. The angle at  $r$  is  $\theta(r) = 3ks/(4\pi r \Gamma) = 3k/(8\pi r \Gamma E_s)$ . (b) A reader of unit sense gains the angle per interval, so its wave number changes by the gradient,  $dk_{\text{dB}}/dt = -\nabla \theta = 3k/(8\pi r^2 \Gamma E_s)$ ; Coulomb's law in nature's units,  $dk_{\text{dB}}/dt = \alpha c/r^2$ , gives  $\alpha_{\text{law}} = \sqrt{3} r^2 dk_{\text{dB}}/dt = (3\sqrt{3}/8\pi) k/(\Gamma E_s) = 0.20675 k/(\Gamma E_s)$ ; the holder's weight enters the source and the read alike as  $k$  and  $E_s$ , and the family's gap cancels. (c) The reader's shift: the step carries  $e^{-i\theta_0}$  on  $z_{\text{next}}$ , so a sense + quantum goes from  $\omega$  to  $\omega - \theta_0$  at its fixed Wronskian (S.33), its energy from  $T \sin \omega$  to  $T \sin(\omega - \theta_0)$ ,  $\delta E = -T\theta_0 \cos \omega$  per quantum; a body of  $n$  quanta and sense  $\pm$  shifts by  $\mp n T \theta_0 \cos \omega = -2\theta_0 \cos \omega W$  with  $W = \pm n T/2$ , the force  $\pm n \nabla \theta$ , Coulomb's  $q_1 q_2$  with  $q = \pm ne$  and  $e = T/2$ ; the reader sources the holder it reads at the same weight, so the pair is reciprocal (S.43 (b)). (d) The number: nature's  $\alpha = 1/137.036$  fixes  $k/(\Gamma E_s) = 1/28.33$  and nothing else;  $k$  is the file's integer read from this one click as  $E_g$  is from  $G$ , and at  $k = 1$  no integer  $\Gamma E_s$  gives it (28 gives  $1/135.43$  and 29 gives  $1/140.27$ ). (d') The energy line. The weight is the product of the write's and the read's,  $k = k_w k_r$ , with the read's  $k_r = 1$  on the integer lattice (a read weight would multiply a level the write already rounded away). A plane record under the turn loses  $2W \cos \omega \Delta \theta$  of form per Node per step, the plane's form at the turned levels, per quantum  $T \cos \omega_s (k_r/\Gamma) \Delta \ell$ ; the holder's own form gains  $\sum_i s_i (\ell_{t+1} - \ell_{t-1})$  per step under the write, per quantum  $(k_w/E_s) \Delta \ell$ , the two factors 2 cancelling; so the loss over the gain is  $\kappa = k_r E_s T \cos \omega_s / (k_w \Gamma)$  at every step and for every motion, an identity checked in floating point at  $-0.996 \pm 0.009$  for  $\kappa = 1$ ,  $-0.498$  at 0.5 and  $-1.993$  at 2, and a charged body and its

light conserve if and only if  $k_r E_s T \cos \omega_s = k_w \Gamma$ , the law's line. Under it, with  $k_r = 1$ ,  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) T \cos \omega_s / \Gamma^2$ , at nature's gap  $T = \Gamma^2/28.33$ ,  $E_s$  a unit of the level and the file one integer fewer; the line is per family through  $\cos \omega_s$ , which in nature is 1 to  $10^{-18}$  for every family, so one read weight serves all; light and every real-line family read no turn and stand outside it. The radiated share of a charged body (S.45 (c)): the time level alone carries half of Larmor's power in form, the odd lines the rest, their coefficient open; no rate is claimed. Two derivations on the identity and its number; the line the owner's decision. (e)  $\Gamma$  is bounded by nothing of the clocks: the content is an integer and the clock reads it as  $p_0 = \Gamma(1 - 1/\Gamma)^c$ , so the finest level at one Node and one interval is  $1/\Gamma$ ; but the write adds  $(\sum wq + r) \text{ div } E$  with the remainder  $r$  kept, so a sub-unit source does not vanish but writes one level once every  $1/s$  intervals, the level at a Node a sequence of 0 and 1 whose mean is the exact rest  $3s/(4\pi r)$ , a first-order sigma-delta modulation; a clock in such a field reads the level 1 for the fraction  $\Delta U \Gamma$  of its intervals and its mean redshift is  $\Delta U$ , finer than  $1/\Gamma$ , and a bound body spanning many Nodes averages their independent dithers, so the gravitational clocks of nature bound the mean potential and not  $\Gamma$ ; nothing bounds  $\Gamma$  from above and no line fixes it:  $\Gamma$  is free and large (two derivations). Check:  $3\sqrt{3}/(8\pi) = 0.20675$ ;  $137.036 \times 0.20675 = 28.33$ ; the implementation's own share formula at [2, 3] and  $\Gamma = 6,000$  gives  $|W|/(\text{count } T) = 0.6708 = 1/(2 \sin \omega_0)$  for a record of count 1 and  $1/2$  for the quantum of the family;  $3\sqrt{3}/(8\pi) \times 28.33 = 1/137.0$ .

**Derivation S.27** (the units, the highest photon energy and the matter pair's gap, Section 10.1). Inputs: the electron as the free quantum of matter, rotating at  $\omega_0$  per interval, and nature's electron at  $m_e c^2/\hbar$  per second; light's band  $\cos \omega = 1 - \frac{1}{3} \sum_a (1 - \cos k_a)$  (S.1); the highest photon observed, 1.4 PeV. Steps: one interval is  $\omega_0 \hbar / (m_e c^2) = \omega_0 \times 1.288 \times 10^{-21}$  s, and one Link, light moving  $1/\sqrt{3}$  Link per interval at  $c$ , is  $\sqrt{3} \omega_0 \bar{\lambda}_e = \omega_0 \times 6.69 \times 10^{-13}$  m with  $\bar{\lambda}_e = \hbar / (m_e c) = 3.862 \times 10^{-13}$  m. The band's top along the body diagonal,  $k_a = \pi$  for all  $a$ , is  $\cos \omega = -1$ ,  $\omega = \pi$ , and along an axis,  $k = \pi$  alone,  $\cos \omega = 1/3$ ,  $\omega = 1.231$ . Along an axis the group velocity vanishes at the edge; along the diagonal,  $k_a = q$  on every axis, the band is  $\omega = q$  exactly, so the speed stays  $1/\sqrt{3}$  up to the corner, where the band has a cusp and no one gradient. So the highest energy a light record carries is  $\pi m_e c^2 / \omega_0 = 1.605 \text{ MeV} / \omega_0$  along the diagonal and  $1.231 m_e c^2 / \omega_0 = 0.629 \text{ MeV} / \omega_0$  along an axis. A photon of 1.4 PeV that arrived, with no orientation of the lattice known, needs the diagonal's ceiling above it,  $1.605 \text{ MeV} / \omega_0 > 1.4 \times 10^9 \text{ MeV}$ ,  $\omega_0 < 1.1 \times 10^{-9}$ . Then  $\text{den} = 2/\omega_0^2 > 1.5 \times 10^{18}$  and  $\gamma_K - 1 = 1/\cos \omega_0 - 1 \approx \omega_0^2/2 < 6.6 \times 10^{-19}$  (S.19);  $\Gamma$  reads nothing of  $\omega_0$ ,  $\alpha$  fixing  $k/(\Gamma E_s)$  and not  $\Gamma$  (S.26). Were the photon's path along an axis, the axis ceiling gives  $\omega_0 < 4.5 \times 10^{-10}$ ,  $\text{den} > 10^{19}$  and  $\gamma_K - 1 < 1.0 \times 10^{-19}$ , a sufficient bound and not a necessary one. Check:  $\hbar / (m_e c^2) = 1.0546 \times 10^{-34} / 8.187 \times 10^{-14} = 1.288 \times 10^{-21}$  s;  $\sqrt{3} \times 3.862 \times 10^{-13} = 6.69 \times 10^{-13}$  m;  $\pi \times 0.511 = 1.605$ ;  $1.231 \times 0.511 = 0.629$ ;  $1.605 / 1.4 \times 10^{-9} = 1.15 \times 10^{-9}$ ;  $[\pi - \omega(\pi - \epsilon)] / (\sqrt{3} \epsilon) = 0.5774$  at  $\epsilon = 10^{-2}$  and  $10^{-3}$  on the diagonal.

**Derivation S.28** (the content's index for every family, Section 8 (p)). Inputs: Eq. (4) at the vacuum's pace and at a content  $U$  under the composed paces,  $(p_0/\Gamma)^2 = N^2$  and  $(p_a/\Gamma)^2 = N^4$  with  $N = (1 - 1/\Gamma)^c = e^{-U}$  to  $O(1/\Gamma)$  (Derivation S.2); a stationary boundary conserves the rotation  $\omega$ . Steps: outside,  $\cos \omega = 1 - g - \frac{\text{num}}{3 \text{den}} (1 - \cos k)$  with  $g = 1 - \text{num}/\text{den}$ ; inside,  $\cos \omega = 1 - gN^2 - \frac{\text{num}}{3 \text{den}} N^4 (1 - \cos k_{\text{in}})$ . Equating,  $\frac{\text{num}}{3 \text{den}} N^4 (1 - \cos k_{\text{in}}) = \frac{\text{num}}{3 \text{den}} (1 - \cos k) + g(1 - N^2)$ , so  $1 - \cos k_{\text{in}} = [1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}] / N^4$ , exact in  $k$  on the domain where the right side lies in  $[0, 2]$ . With  $1 - N^2 = 2U - 2U^2 + \dots$  and  $N^{-4} = 1 + 4U + \dots$  the first order is  $[1 - \cos k + 6(\text{den} - \text{num})U/\text{num}] / (1 - 2U)^2$ , the quadratic forms' line, the mass term's second order  $-6(\text{den} - \text{num})U^2/\text{num}$ . For light,  $g = 0$ :  $1 - \cos k_{\text{in}} = (1 - \cos k)/N^4$ , so  $\sin(k_{\text{in}}/2) = \sin(k/2)/s$  with  $s = N^2$ , exact in the law's variable  $2 \sin(k/2)$  where  $\sin(k/2) \leq s$ . The phase index is  $k_{\text{in}}/k = (2/k) \arcsin[\sin(k/2)/s] = 1/s + k^2(1/s^3 - 1/s)/24 + O(k^4)$ , which is  $1/s = e^{2U}$ , Derivation S.2's long-wavelength index, at  $k \rightarrow 0$  alone. At  $k = \pi/2$  and  $s = 0.8$  the exact ratio is 1.380 against  $1/s = 1.25$ , and to first order in  $U$  the index is  $1 + [4 \tan(k/2)/k] U$ , the coefficient 2 at  $k \rightarrow 0$  and  $8/\pi$  at  $k = \pi/2$ : the lattice's own chromatic dispersion, absent in the ray limit in which the bending, the delay and the lens are read. For slow matter,  $k, k_{\text{in}} \ll 1$ :  $k_{\text{in}}^2 = [k^2 + 6(\text{den} - \text{num})(1 - N^2)/\text{num}] / N^4$ , so  $n_m^2 = k_{\text{in}}^2/k^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)] / N^4$ , to first order  $[1 + 12(\text{den} - \text{num})U/(\text{num} k^2)] / (1 - 2U)^2$ . Check: with  $k^2 = m^{*2} v^2$  and  $m^* = 3 \tan \omega_0$  (S.22), the  $1/k^2$  term is  $12(\text{den} - \text{num})U/(\text{num} m^{*2} v^2) = [4(1 - \cos \omega_0)/(3 \cos \omega_0 \tan^2 \omega_0)] U/v^2$ , which as  $\omega_0 \rightarrow 0$  is  $(2/3) U/v^2 = 2(U/3)/v^2 = 2\Phi/v^2$  with  $\Phi = c^2 U$ ,  $c^2 = 1/3$ : nature's index for a slow body in a potential. The factor  $6(\text{den} - \text{num})/\text{num}$  over  $m^{*2}$  is the matter band's own, the Kepler column of Eq. (14), and  $1/N^4$  the Link's double read.

**Derivation S.29** (the clusters' cosmology: the ball of dust, Section 7.5). Inputs: Newton's pull between bodies at Kepler's  $G_K$  (Section 8 (j), Derivation S.25); a finite ball of dust of radius  $a(t)$  and uniform density  $\rho$  on the static lattice; the Doppler period of a receding emitter, row (m), and the moving clock's factor, row (e). Steps: a shell at  $a$  feels the mass inside it,  $M(a) = (4\pi/3)\rho a^3$ , and nothing from outside (Newton's theorem on a sphere, which holds for the  $1/r$  field of Section 8 (j)), so  $\ddot{a} = -G_K M(a)/a^2 = -(4\pi G_K/3)\rho a$ . Multiplying by  $\dot{a}$  and integrating with  $\rho a^3$  constant gives  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho - \kappa/a^2$  with  $\kappa$  the integration constant, positive where the ball is bound: Friedmann's two equations for dust, as Milne and McCrea derived them for the Newtonian ball. The deceleration parameter is  $q_0 = -\ddot{a} a / \dot{a}^2 = (4\pi G_K \rho/3)/H^2 = \Omega_m/2$  with  $\Omega_m = 8\pi G_K \rho/(3H^2)$ , positive for every  $\rho > 0$ . The redshift: the period of

a receding emitter at the speed  $\beta c$  is read  $(1+\beta)$  times longer at the resting detector (row (m)) and the emitter's own tick is longer by the moving clock's factor  $1/\sqrt{1-\beta^2}$  at  $c_m = c$  (row (e)), so  $1+z = (1+\beta)/\sqrt{1-\beta^2} = 1+\beta+\beta^2/2+\dots$ , nature's relativistic Doppler form to the second order, the classical  $1/(1-\beta) = 1+\beta+\beta^2$  differing at the second order. At the computed pair  $c_m = 0.867c$  the second order differs, as every number of the computed pair does. Every click's duration is stretched by the same  $1+z$ , the light curves' time dilation, and a surface's brightness falls by  $(1+z)^{-4}$ , Tolman's dimming, both from the same two rows; light from a source at  $r$  falls into the ball's deeper centre, a blueshift  $\Delta U = (\Omega_m/4)(H_0 r/c)^2$  beside the Doppler, the second order that makes  $d_L(z)$  Friedmann's with  $q_0$  unchanged. The vacuum's write: a uniform source  $s$  in the massless row's rest,  $\nabla^2 L = 3s$  in the lattice's units ( $c^2 = 1/3$ ), has the solution  $L = L_{\text{faces}} - s(R^2 - r^2)/2$  on the ball, a bowl rising outward. Its pull on a shell at  $a$  is  $+\Lambda c^2 a/3$  with  $\Lambda$  proportional to the write's rate  $|q_{\text{vac}}|/(E_v \Gamma L W)$ , so  $\ddot{a} = -(4\pi G_K/3)\rho a + \Lambda c^2 a/3$  and  $\dot{a}^2/a^2 = (8\pi G_K/3)\rho + \Lambda c^2/3 - \kappa/a^2$ , Friedmann's equations with a cosmological constant;  $q_0 = \Omega_m/2 - \Omega_\Lambda$ , and  $\ddot{a} = 0$  where  $\rho = 2\rho_\Lambda$ ,  $1+z_t = (2\Omega_\Lambda/\Omega_m)^{1/3}$ ,  $z_t = 0.67$  at  $\Omega_\Lambda = 0.7$ . Light's band at  $(\pi, \pi, \pi)$  needs  $L \geq 0$  at every Node (the guard), so  $L_{\text{faces}} \geq sR^2/2$ , in  $U$  the faces' content  $\Lambda R^2/6$ ; at one astronomical unit the bowl's pull  $\Lambda r/3$  is  $2 \times 10^{-22}$  of the Sun's.

**Derivation S.30** (the shadow's capture radius from light's index, Section 8). Inputs: light's index  $n(r) = e^{2U}$  with  $U = m/r$  (Derivation S.2 under the composed paces),  $m = GM/c^2$  in Links. Steps: in a static isotropic index a ray at the impact parameter  $b$  keeps  $n(r)r \sin \psi = b$  along its path, so a circular ray stands where  $d(nr)/dr = 0$ :  $d(re^{2m/r})/dr = e^{2m/r}(1 - 2m/r) = 0$  at  $r = 2m$ , the photon sphere, and the capture radius is  $b = n(2m)2m = 2em = 5.437m$ . Schwarzschild's is  $3\sqrt{3}m = 5.196m$ , so the shadow is larger by  $2e/(3\sqrt{3}) - 1 = 0.046$ . Check: with the quadratic index  $1/(1-2U)$  the same condition gives  $r = 4m$  and  $b = 8m$ , +54 percent, which the Event Horizon Telescope excludes; the composed index is the one the principle gives.

**Derivation S.31** (the two clocks under the principle, Section 10.2). Inputs: the principle of Section 3.4, a uniform level  $U$  a change of units, time by  $N$  and length by  $h$  with  $hN = 1$ ; light's speed  $N^2$  (S.2). Steps: a bound body in the level is the flat body in the local units, so its size is  $1/h = N$  Links. Its band at the Node is  $2 - 2\cos \omega = 2(1-\nu)N^2 + (\nu/3)N^4 \sum_a s(k_a)^2$  with  $\nu = \text{num}/\text{den}$  and  $s(x) = 2\sin(x/2)$ , since  $(p_0/\Gamma)^2 = N^2$  and  $(p_a/\Gamma)^2 = N^4$ , and its lattice wave numbers are  $k = k_{\text{proper}}/N$ , so  $(\nu/3)N^4 \sum_a s(k_a)^2 = (\nu/3)N^2 \sum_a k_{\text{proper},a}^2 [1 - k_a^2/12 + \dots]$ : every mode of the record, the rest rotation and every level of its spectrum, scales by  $N$  in  $s(\omega)$ , exactly at long wavelength and to the lattice's  $k^2/12$  at finite  $k$ . In  $\omega$  the rest line's rate is  $-2\tan(\omega_0/2)/\omega_0 = -f(\omega_0)$  per unit  $U$ , Derivation S.15's number. A difference  $s(\omega'_2) - s(\omega'_1)$  scales by  $N$  exactly and is no observable. The temporal beat a detector reads,  $\Delta' = \omega'_2 - \omega'_1$  in  $\cos(\Delta't)$ , with  $\omega' = 2\arcsin[N\sin(\omega/2)] = N\omega[1 - (1-N^2)\omega^2/24 + O(\omega^4)]$ , is  $\Delta' = N\Delta[1 - (1-N^2)(\omega_1^2 + \omega_1\omega_2 + \omega_2^2)/24 + O(\omega^4)]$ : at  $N = 0.8$ ,  $\omega_1 = 0.4$  and  $\omega_2 = 0.9$  the exact beat is 0.3916 against  $N\Delta = 0.4$  and the series 0.3920. The relative correction is  $2.5 \times 10^{-38}$  at nature's gap at the axis bound ( $\omega$  of  $10^{-14}$ ,  $U = 10^{-9}$ ;  $10^{-19}$  at the LHAASO bound) and 1.4 percent at the implementation's  $\omega_b = 0.78$  and  $U = 0.1$ , the lattice's arithmetic and not the law's, which the implementation's prediction reads in both variables. A cavity whose mirrors are bodies stands  $L/h = LN$  Links apart and its round trip is  $2LN\sqrt{3}/N^2 = 2L\sqrt{3}/N$  intervals, the rate  $N$  exactly with no  $\omega$  correction, the atom's beat to its correction  $(1-N^2)\omega^2/24$  above; a cavity of declared faces keeps its  $L$  Links and ticks at  $N^2$ . So under the principle no clock type is singled out, which is what the null redshift tests measure, while a cavity of declared faces is an instrument's Node and no body of nature.

**Derivation S.32** (the conserved form's Link factor, two witnesses, Theorem 3). Inputs: Theorem 3's line without its division on a periodic board at the content 0 with every Link's factor  $q$ , so  $p_0 = p_i = \Gamma$ ,  $R = 2\text{num } q^2$ ,  $w = 6\text{den } \Gamma^2$  and  $S = 12\text{num } \Gamma^2 - 6R$ ; the form  $E$  of Eq. (6) and the same form with the plain Link term  $-2\text{num } \sum_i \text{now}_i S_6(\text{before})_i$ ; exact fractions, no run of the implementation. First witness: three Nodes along  $x$  with  $y$  and  $z$  folded,  $\text{num} = \text{den} = 1$ ,  $\Gamma = 4$ ,  $q = 2$  ( $R = 8$ ,  $w = 96$ ,  $S = 144$ ),  $\text{now} = (1, 0, 0)$  and  $\text{before} = 0$ : one step gives  $\text{next} = (11/6, 1/12, 1/12)$ ; the plain form goes from 6 to  $-21/4$  and Eq. (6) stays 6. Second witness:  $3^3$  Nodes at  $[2, 3]$ ,  $\Gamma = 10$ ,  $q = 5$  ( $R = 100$ ,  $w = 1,800$ ,  $S = 1,800$ ), a uniform  $\text{now} = 3$  over  $\text{before} = 0$ : the step gives  $\text{next} = 4$  at every Node with the remainder 0, so the integer step and the line agree; the plain form per Node goes from 162 to  $-54$  and Eq. (6) stays 162 (4,374 over the board). Both are an independent automated check's, recomputed here. Check: at  $q = \Gamma$  the factor is 1 and the two forms coincide, the domain of the plain current.

**Derivation S.33** (magnetism: the continuity identity and the covector, Section 8). Inputs: a plane record  $z = \text{re} + i \text{im}$  stepping by Rule3 under the holder of the sign (Section 5.5):  $e^{-i\theta_0} z_{\text{next}} + e^{i\theta_0} z_{\text{before}} = [S z_{\text{now}} + \sum_a R_a (e^{i\theta_a} z_{+a} + e^{-i\theta_a} z_{-a})]/w$ , with  $\theta_0$  the time turn and  $\theta_a$  the phase on the axis  $a$ 's Links, read at the interval's start; the Wronskian with the angle,  $W_i = \text{Im}(\bar{z}_{i,\text{now}} e^{i\theta_0} z_{i,\text{before}})$ ; the Link quantity  $g_{ij} = \text{Im}(\bar{z}_i e^{i\theta_{ij}} z_j)$ , antisymmetric,  $g_{ji} = -g_{ij}$ , since the phase is read  $+\theta_a$  through the  $+a$  Port and  $-\theta_a$  through the  $-a$  Port. Steps: (a) The identity, with the angle fixed over the step (when a level changes in time the booked  $W$  moves with the record's amplitude by  $\Delta W/W = \mp \cot \omega \Delta \theta$ , nothing at nature's  $10^{-13}$ , so the charge is exact for static levels and to first order in a level's change). Multiply the step by  $w \bar{z}_{\text{now}}$  and take the imaginary part; the  $S$  term is real and drops:  $w \text{Im}(\bar{z}_{\text{now}} e^{-i\theta_0} z_{\text{next}}) + w \text{Im}(\bar{z}_{\text{now}} e^{i\theta_0} z_{\text{before}}) =$

$\sum_a R_a (g_{i,i+a} + g_{i,-a})$ . The first term is  $-W_i(t+1)$  and the second  $W_i(t)$ , and  $g_{i,-a} = -g_{-a,i}$ , so

$$w [W_i(t+1) - W_i(t)] + \sum_a [R_a g_{i,i+a} - R_a g_{i,-a}] = 0 :$$

an exact lattice continuity equation at every Node and interval, the density  $\rho_i = w W_i$  and the flux through the Link  $(i, i+a)$  equal to  $R_a g_{i,i+a}$ , for any paces, any phases and any levels; the integer step's remainders add the term of S.9's domain. (b) Flux over density is the group velocity. For  $z = A e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)}$ ,  $\rho = w A^2 \sin \omega$  and the flux is  $R_a A^2 \sin k_a$  (common sign), so flux over density is  $(R_a/w) \sin k_a / \sin \omega = d\omega/dk_a$  from the band  $2w \cos \omega = S + 2 \sum_a R_a \cos k_a$ : the current is  $\rho v$  exactly, and the law's Node current  $J_a = g_{i,i+a} + g_{i,-a}$  obeys  $J_a = (6 \text{ den/num}) W v$  on a plane record, nature's  $j = \rho v$  with the band's constant and no  $\sin \omega_b$ . (c) The covector. Inserting the plane wave in the step with the phases gives  $\omega = \theta_0 + \omega_{\text{band}}(\mathbf{k} + \boldsymbol{\theta})$ : the pair  $(\theta_0, \theta_a)$  shifts  $(\omega, k_a)$  and transforms as  $(\omega, \mathbf{k})$  does; at long wavelength every band is the Lorentz form  $\omega^2 = \omega_0^2 N^2 + c_s^2 k^2$  with  $c_s^2 = \text{num}_s / (3 \text{ den}_s)$  (from  $2 \cos \omega = S/w + (2 \text{ num}/3 \text{ den}) \sum_a \cos k_a$ ), so  $(\theta_0, \theta_a)$  is a covector under the band's boost: minimal coupling,  $A_0$  on the time step and  $A_a$  on the Link. (d) The halved current over the time level's wall. Maxwell's pair with one kernel sources  $(\theta_0, \theta_a)$  from  $(\rho, \text{flux}/c^2)$  over one wall. The Node's flux on the axis  $a$  is the mean of its two Links',  $R_a(g_{i,i+a} + g_{i,-a})/2 = R_a J_a/2$ , and  $R_a/w = \text{num}/(3 \text{ den})$  at the vacuum's pace, so flux over  $c_s^2 \rho$  is  $[(\text{num}/3 \text{ den}) J_a/2] / [(\text{num}/3 \text{ den}) W] = (J_a/2)/W$ : the odd line on the axis  $a$  is sourced by the halved current  $J_a/2$  over the same wall as the time level's  $W$ , with no number, when the pair is read at the source family's own  $c_s$ ; read at light's  $c^2 = 1/3$  the factor  $c_s^2/c^2 = \text{num}_s/\text{den}_s$  appears, 1 for light,  $1 - 10^{-28}$  at nature's matter,  $2/3$  at [2, 3]. The halving is not a choice: the Node's current is the mean of its two Links as the continuity identity defines it, as the tension on an axis is the mean of its two stresses. (e) Coulomb over  $\gamma$ . For a source moving at  $v$  along the Link and a reader co-moving with it, the time level is the moving density's,  $\gamma$  times the rest level (the record contracted at fixed total  $W$ ), so the electric force across the beam is  $\gamma F_0$ ; the odd level is sourced by  $J_a/2 = W v$ , the reader at  $v$  feels  $-v \partial \theta_a / \partial y$ , a magnetic force  $\beta^2 \gamma F_0$  toward the source; the total is  $\gamma(1 - \beta^2) F_0 = F_0/\gamma$ . Check:  $1/\gamma = 0.99499$  at  $\beta = 0.1$  and  $0.86603$  at  $\beta = 0.5$ , the row (b2)'s prediction; the identity (a) holds in exact arithmetic on a chain with contents and phases at every Node (the same check as S.9's, the phases a unitary on each read). Domain: the covector is the band's at long wavelength and breaks at the lattice's anisotropy  $k^2/12$  ( $2 \times 10^{-4}$  at  $k = 0.05$ ,  $10^{-29}$  at nature's Compton wavelength); the pairing holds at the source family's own  $c_s$ , so a source and a reader of one family see one light speed.

**Derivation S.34** (the composed paces, the acts' factors and the exponential metric, Section 3.4). Inputs: the model's one premise beside the line, that a level is a count of units and every act is measured in the Node's own units (Section 3.4); the clock  $p_0$  as Rule3's coefficient in  $S$ ; the band at a pace, S.1; the one assumption  $p_a = p_0^2/\Gamma$ . Steps: (a) The composition. One unit of level is one act on the clock as it stands, so it slows the clock by one part in  $\Gamma$  of the clock as it stands:  $p_0(c+1) = p_0(c)(1 - 1/\Gamma)$ , whence  $p_0(c) = \Gamma(1 - 1/\Gamma)^c$ . The functional equation  $P(U+l) = P(U)P(l)$  is Cauchy's, whose continuous solutions are the exponentials and no other, so  $N = p_0/\Gamma = e^{-U}(1 + O(U/\Gamma))$  with  $U = c/\Gamma$ ; the quadratic form's subtraction  $p \rightarrow p - 2$  is as memoryless and is excluded only by the premise that the read is an act measured in the cell's units, like the write. Check: at  $\Gamma = 6,000$  and  $c = 300$ ,  $(1 - 1/\Gamma)^c = 0.951225$  against  $e^{-0.05} = 0.951229$ , a relative  $4 \times 10^{-6}$ , which is  $U/(2\Gamma)$ . (b) The ruler.  $p_a = p_0^2/\Gamma = \Gamma N^2$  gives the axis ruler  $h = p_0/p_a = 1/N$ :  $hN = 1$ . (c) The metric. S.1 at the paces reads  $2(1 - \cos \omega) = 2(1 - \nu)N^2 + (\nu/3)N^4 \sum_a s(k_a)^2$  with  $s(x) = 2 \sin(x/2)$ ; at long wavelength  $\omega^2 = N^2 [\omega_0^2 + c_m^2 (k/h)^2]$  with  $k/h = kN$ : Klein-Gordon's on a static metric with the lapse  $N$  and the spatial scale  $h$ ,  $ds^2 = -N^2 dt^2 + h^2 d\mathbf{x}^2 = -e^{-2U} dt^2 + e^{2U} d\mathbf{x}^2$ , the exponential metric in isotropic coordinates (Yilmaz 1958), whose post-Newtonian expansion is  $g_{00} = -(1 - 2U + 2U^2)$  and  $g_{ij} = (1 + 2U + 2U^2)\delta_{ij}$ :  $\beta = \gamma = 1$ , the spatial second order 2 against Schwarzschild's 1.5 (S.30). (d) The acts' factors. Rule3 measured in the cell's units: a count per Node is a count per proper volume  $h^3$ , a step per interval is a step per proper interval  $N$ , and the write is Rule3's second difference in time, so a count source carries  $N^2/h^3 = N^5 = p_x p_y p_z / (p_0 \Gamma^2)$  (check:  $\Gamma^3 N^6 / (\Gamma N \cdot \Gamma^2) = N^5$ ); the Wronskian, one time difference, carries  $N/h^3 = N^4 = p_x p_y p_z / (p_0^2 \Gamma)$  and its readers' turn the remaining  $N$ ,  $\theta = (\text{level}/\Gamma)(p_0/\Gamma)$ , the turn per proper interval; the current's one space difference gives  $h^4$  in coordinates against the Peierls phase's  $h$  per coordinate Link, the two cancelling, so the Link angle is plain. With a tension the three rulers differ,  $h_a = p_0/p_a$ , and the volume is  $h_x h_y h_z$ . (e) The truncation. The clock's quadratic form  $(\Gamma - c)^2 + c^2 = \Gamma^2(1 - 2U + 2U^2)$  is  $\Gamma^2 e^{-2U} = \Gamma^2(1 - 2U + 2U^2 - 4U^3/3 + \dots)$  to the second order, differing at  $-4U^3/3$ ; the Link's  $(\Gamma - 2c)^2 = \Gamma^2(1 - 4U + 4U^2)$  is  $\Gamma^2 e^{-4U} = \Gamma^2(1 - 4U + 8U^2 - \dots)$  to the first order only. Check: at  $U = 0.05$  the clock's form gives 0.905000 against  $e^{-0.1} = 0.904837$ , the difference  $1.63 \times 10^{-4}$  against  $4U^3/3 = 1.67 \times 10^{-4}$ ; the Link's gives 0.8100 against  $e^{-0.2} = 0.8187$ . The horizon of the quadratic forms at  $c = \Gamma/2$  is the truncation's zero of  $\Gamma - 2c$  and not the law's (S.3).

**Derivation S.35** (the preferred-frame parameter of a diagonal metric, Section 10.3). Inputs: the model's metric is diagonal in the lattice's frame by construction: the content enters the clock  $p_0$  and the Link pace  $p_a$ , both real and even in time and in space, the content's rest is Poisson's in the lattice's frame with no speed in its kernel, and

no odd real Link read exists, since a real asymmetric hopping with a curl breaks the reciprocity the conserved form needs (Theorem 3); so  $g_{0a} \equiv 0$  in the lattice's frame for every motion of the source (Section 10.3). The parametrised post-Newtonian metric in the theory's preferred frame (Will's review, the paper's reference, in the standard gauge):  $g_{0j} = -\frac{1}{2}(4\gamma + 3 + \alpha_1 - \alpha_2 + \zeta_1 - 2\xi) V_j - \frac{1}{2}(1 + \alpha_2 - \zeta_1 + 2\xi) W_j$ , with  $V_j$  and  $W_j$  the two vector potentials of the matter current, related by  $V_j - W_j = \chi_{,0j}$  for the superpotential  $\chi$ . Steps: (a) The post-Newtonian gauge freedom  $x^0 \rightarrow x^0 + \lambda \chi_{,0}$  adds  $\lambda \chi_{,0j} = \lambda(V_j - W_j)$  to  $g_{0j}$ , moving weight between  $V_j$  and  $W_j$  and leaving the sum of their two coefficients,  $(4\gamma + 3 + \alpha_1 - \alpha_2 + \zeta_1 - 2\xi) + (1 + \alpha_2 - \zeta_1 + 2\xi) = 4\gamma + 4 + \alpha_1$ , gauge-invariant. (b) A metric whose  $g_{0j}$  vanishes in its preferred frame for every source current has both coefficients 0 in some gauge, so the invariant sum vanishes:  $4\gamma + 4 + \alpha_1 = 0$ ,  $\alpha_1 = -4(1 + \gamma)$ , which is  $-8$  at the model's  $\gamma = 1$  (S.19 in  $U$ ). (c) The frame-dragging precession of a gyroscope near a rotating body is  $(4\gamma + 4 + \alpha_1)/8$  of general relativity's Lense-Thirring value, 1 in general relativity and 0 in the model: Gravity Probe B's 0 against  $37.2 \pm 7.2$  milliarcseconds per year is the same fact as  $\alpha_1 = -8$ , read by two instruments. (d) Had the model's read been covariant, the boost of its static metric  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$  to the velocity  $v$  would give  $g_{0i} = \gamma^2 v_i (N^2 - h^2) = -4U v_i$  to first order, Einstein's, with no new coefficient; the diagonal read in the lattice's frame is what removes it, so the finding is the frame's and not a number's (the carrier's theorem, S.36). Check:  $-4.000$  for the boosted cross term at first order; nature's  $\alpha_1$  from the binary pulsar PSR J1738+0333 is  $|\alpha_1| < 4 \times 10^{-5}$  (Shao and Wex 2012) and from lunar laser ranging  $10^{-4}$ , so the finding stands at  $2 \times 10^5$  and  $10^5$ ; Gravity Probe B's drag at 5 standard deviations; the two are one finding, not two. Domain:  $\alpha_1$  alone is gauge-invariant here;  $\alpha_2$  and the velocity-dependent terms of  $g_{00}$  depend on how a moving record's count writes, which the is not settled, so the paper states  $\alpha_1 = -8$  and the drag 0 and no more.

**Derivation S.36** (the carrier's theorem, Section 10.3). Inputs:  $g_{0a}$  is the term of the wave operator odd in time and odd in space together,  $2g^{0a}\partial_t\partial_a$ ; Rule3's step is explicit (next from now and before at the Node and its six neighbours), of one Link's radius, bijective (Theorem 1), and conserves the form (Theorem 3), whose proof needs the Link coefficients reciprocal; a real record has no phase; the dispersion of a term is read by inserting the plane wave  $a = A \cos(kx - \omega t)$ . Steps, the four ways on the synchronous clock: (i) The one-sided real mixed term  $C_a[(a_{+a} - a_{-a})_{\text{now}} - (a_{+a} - a_{-a})_{\text{before}}]$  added to the line gives  $w(e^{-i\omega} + e^{i\omega}) = S + 2 \sum_a R_a \cos k_a + C_a(2i \sin k_a)(1 - e^{i\omega})$ , and  $1 - e^{i\omega} = -2i \sin(\omega/2) e^{i\omega/2}$ , so the term is  $4C_a \sin k_a \sin(\omega/2)[\cos(\omega/2) + i \sin(\omega/2)] = 2C_a \sin k_a \sin \omega + 4iC_a \sin k_a \sin^2(\omega/2)$ : the  $g_{0a}$  part beside an imaginary part, a growth or decay of  $\omega$  that breaks the form and the exact inverse. Check: at  $\Gamma = 6,000$ , light's row,  $k = 0.3$ ,  $C/w = 10^{-2}$ :  $\omega = 0.16984 - 2.47 \times 10^{-4}i$  against the band's  $0.17277$  (the first-order prediction  $-2.56 \times 10^{-4}i$ ). (ii) The time-centred mixed term, with the neighbours' next, is conservative but implicit: it needs the neighbours' next, outside the one-Link dependency radius and the explicit inverse. (iii) A real asymmetric hopping with a curl,  $R_{i \rightarrow j}/R_{j \rightarrow i}$  not a ratio of Node weights, has no symmetric step kernel, so Theorem 3 fails. (iv) A Peierls phase on the Link carries  $g_{0a}$  for a plane record only, a real record having no phase, and it shifts  $k$  by the same angle for every record, a force independent of  $\omega_0$ : it couples to the charge and not to the mass; light and neutral matter are real lines. So no real, local, explicit, bijective line on the synchronous clock carries  $g_{0a}$ . The fifth way, and why it fails: clock the cube's two sublattices alternately, half an interval apart, each Node reading its neighbours in the two slices that bracket its own time, their mean into the hopping and their difference into an odd Link level  $C_{ij}$ :  $w(a_i(t+1) + a_i(t-1)) = S a_i(t) + \sum_j R_{ij}[a_j(t + \frac{1}{2}) + a_j(t - \frac{1}{2})]/2 + \sum_j s_{ij} C_{ij}[a_j(t + \frac{1}{2}) - a_j(t - \frac{1}{2})]$ . Its dispersion,  $2w \cos \omega = S + 2 \cos(\omega/2) \sum_a R_a \cos k_a + 4 \sum_a C_a \sin \omega \sin k_a$ , is real for every  $k$ , the step's eigenvalues lie on the unit circle (to  $10^{-15}$  on a ring of 12 Nodes,  $C$  uniform or a random Link field), the mixed term is exactly  $g_{0a}$ , and it is real, local, explicit and bijective. But a uniform record sampled at the neighbours' half-slices is  $a(t) \cos(\omega/2)$ , so the hopping does not cancel at  $k = 0$  and the rest rotation in a content becomes  $1 - \cos \omega = (1 - \nu)N^2 + \nu N^4(1 - \cos(\omega/2))$  in place of  $(1 - \nu)N^2$ ; at small  $\omega$ ,  $\omega^2 = 8(1 - \nu)N^2/(4 - \nu N^4)$ , and for  $\nu \rightarrow 1$ ,  $\omega/\omega_0 = N\sqrt{3/(4 - N^4)} = N(1 - 2U/3 + \dots)$ : a clock in a well runs at  $\omega_0(1 - 5U/3)$ , a redshift  $5/3$  of general relativity's. Check:  $\sqrt{3/(4 - 0.9^4)} = 0.9472$ , the three-dimensional staggered band's  $0.948$  at  $N = 0.9$ ,  $\nu = 0.99$  on a  $13^3$  grid, its light speed  $c^2 = 4/9$  and its top  $2\pi/3$ ; Pound and Snider's 1 percent (1965) excludes the 67 percent at 60 standard deviations, Gravity Probe A's  $1.4 \times 10^{-4}$  at  $5 \times 10^3$ . Demanding the exact redshift for every small-gap family,  $P_0(4 - \nu) = N^2(4 - \nu P_a)$  for all  $\nu$ , forces  $P_0 = N^2$  and  $P_a = 1$ : the Link may not read the content, light's coordinate speed  $1/N$ ,  $\gamma_{\text{PPN}} = -2$ . So the fifth way fails the clock as the first three fail the inverse, the form or the radius and the fourth the real record, and the theorem stands on Rule3's clock:  $\alpha_1 = -8$  and the drag 0 are structural (S.35), and the carrier remains a hypothesis by name.

**Derivation S.37** (confinement as the indivisibility of a record, Section 10.3). Inputs: a family of dimension 3: three real lines at every Node stepped by the one line, with a rotation among the three on each Link (the colour holder's phases), as the plane is two lines with a rotation in its plane; the detector's definition (Section 5.3): a click is one whole quantum of a record credited to a declared region over its window, drawn by the share  $Q_R/W_c$  of the inflow through the region's front; the share of a record of several lines is the sum of the lines' forms (Theorem 3 line by line, the Link rotation a unitary on the triple, so the step kernel stays symmetric and the summed form is conserved); the readings of Eq. (1) are the record's, never one line's: the plane's re and im are booked together in  $D$ ,  $F$  and  $W$ , and so are the three lines. Steps: (a) No click reports one line. The click's object is the record's quantum, the share summed over the

lines; no reading of the law credits a line apart, and the credit draws regions, not lines (Section 5.3, "no click names a Node" and the one form of the click). So a record of dimension 3 clicks whole or not at all: no detector ever reports one colour line, as none reports the plane's real part alone. This is confinement with no coupling and no number: the quark model's "no free quark" is the dimension's indivisibility under the click. (b) The partons. A hard click, a scattering read at the momentum transfer  $q$ , reads the record's current at the wave number  $q$  through the region's front; the lines' forms sum to the record's, so what it reads is three fractions summing to one, the lines' shares: the quark model's three is the dimension's three, and the fractions are the record's lay and its Link rotations. (c) Scaling. The levels are deterministic and the count is conserved, so no pair is created and no coupling runs (Section 10.3, the three roots); the three fractions therefore do not change with  $q$ : Bjorken scaling is exact on the GameBoard, where nature reads logarithmic violations of 10 to 20 percent across the measured decades of  $q^2$  (SLAC to HERA), an explicit finding with the same root as  $\alpha$ 's running. (d) The colourless bodies. The holder's sources are the three lines' Wronskian-like currents; a body whose three senses sum to 0 writes no colour level outside itself, the colourless states, and a pair of opposite sense or three coplanar axes are the two ways to 0. (e) What it is not. The binding between two nucleons is between separable records (a nucleus parts into them) and needs the total form to fall at a kept count, which Theorem 3 forbids at fixed paces (S.38); so the word "strong force" parts in two, the confinement a theorem of the click and the nuclear binding open. Check: the conserved form under the Link rotations was verified in exact arithmetic on a chain with random rotations (the step kernel symmetric, the form's drift 0); the credit's unit is the check's own definition, read in every checked world. Status: a theorem of the click's one form, two independent derivations; clicks (no free quark, the partons' fractions); the scaling a finding by name, one derivation.

**Derivation S.38** (the mass defect at a kept count and the binding bound, Section 10.3). Inputs: Theorem 3, the form conserved by Rule3 at fixed paces for any records; the invariant of S.14, a body of  $M$  units of the invariant at the rotation  $\omega_b$  having the share  $M \sin \omega_b$  (the energy per quantum  $\sin \omega_b$  in units of  $T$ , Planck's line); the rest rotation in a content,  $\omega_0(U) = N\omega_0$  to the order  $\omega_0^2$  (S.15 with  $f \rightarrow 1$ ); the massless kernel of a holder at the weight  $W$ : the level at  $r$  Links from a count  $C$  is  $3CW G(r)/E$  with  $G$  the lattice Green's function per unit source,  $G(0) = 0.2527$  (Watson) and  $3/(4\pi r)$  far,  $e^{-r/R}$  beyond a gapped holder's range (S.11). Steps: (a) No energy from combinatorics. Two records that approach, overlap, exchange quanta through their tails, join into one standing record or part again are records stepped by Rule3; if the paces do not change, the total form before and after is the same number to the last bit (Theorem 3), and a click moves share from one record to another and conserves the total too. So no joining, parting or crediting of clusters lowers the share at a kept count: there is no energy released, and a nucleus bound that way would have no binding energy. (b) What the defect is. The one act that changes a body's share at a kept count is a change of its paces, a content read; a body in its own content  $x_{\text{self}}$  rotates at  $N\omega_0$  with  $N = e^{-x_{\text{self}}}$ , so its click mass is  $M \sin(N\omega_0) = M \sin \omega_0 - M\omega_0 x_{\text{self}} \cos \omega_0 + O(x^2)$ : the mass defect is  $M\omega_0 x_{\text{self}}$  to the order  $\omega_0^2$ , the self-read's redshift of the body's own rotation and nothing else; this is the law's "a body that binds is lighter in every field it sources by its binding". Check: the law's standing mode in a well deepened from 300 to 600 levels at  $\Gamma = 6,000$ : the share fell 4.0 percent with the invariant kept to  $10^{-5}$  (Section 7.4). (c) The bound. A body of count  $C$  over  $r$  Links, bound by a holder at the weight  $W$  and the divisor  $E$ , writes on itself at most the level of its whole count at the distance  $r$ ,  $3CW e^{-r/R}/(4\pi r E)$ , so  $x_{\text{self}} \leq 3CW/(\Gamma E)$  and

$$\frac{\text{defect}}{\text{rest}} \times r \leq 0.24 \frac{CW}{\Gamma E},$$

and for a body of one Node  $x_{\text{self}} = 3G(0)CW/(\Gamma E) = 0.758CW/(\Gamma E)$ , the lattice kernel at the source. (d) The numbers. The deuteron's defect is  $2.22/1877.6 = 1.2 \times 10^{-3}$  and the nucleon's in iron  $8.8/939 = 9 \times 10^{-3}$ , so they need  $CW/(\Gamma E) \geq 5 \times 10^{-3}$  and  $4 \times 10^{-2}$ ; at  $r$  about  $10^{10}$  Links for a nucleon (the MAGIC Link) and a nucleon-count body of 1,836 quanta this is  $W$  of about  $2 \times 10^{19}$  for the nucleon's defect and  $10^{21}$  to  $10^{22}$  for a binding of order one at  $\Gamma = 10^{14}$ , taken for illustration, growing with  $\Gamma$ , which no click bounds (S.26); the bound's form does not move with the reading of  $\Gamma$ , its numbers do. The credit's quantum does not supply it: the deuteron's 2.22 MeV is 4.35 quanta of 0.511 MeV and the nucleon's 8 MeV 15.7, not wholes, and no act credits share to nowhere. (e) The one lattice lightening without content, the band's  $\sin \omega_b$  falling to 0 at the band's top, acts at rotations of order one and makes a body nearly massless, not 99 percent of its quanta's mass: not a nuclear binding. Check of the kernel: Watson's integral gives  $G(0) = 0.252731$ ,  $3G(0) = 0.7582$ ;  $G(1) = G(0) - 1/6 = 0.08607$ ,  $3G(1) = 0.2582$ , 1.08 of the far kernel  $3/(4\pi) = 0.2387$  at  $r = 1$ .

**Derivation S.39** (the gapped holder's kernel, Section 10.3). Inputs: a held row's static line in the vacuum, from the start's condition (Section 3.5):  $6 \text{ den } L - \text{num } S_6(L) = 3 \text{ den } \sigma$ , that is  $\text{num } \Delta L - 6(\text{den} - \text{num})L = -3 \text{ den } \sigma$  with  $\Delta = S_6 - 6$  the lattice Laplacian and  $\sigma$  the source per interval; the reach  $\kappa^2 = 6(\text{den} - \text{num})/\text{num}$  of S.11; the write and the read of every rotation holder the same act,  $W$  over  $E_s T$  into the level and the level over  $\Gamma$  into the angle (Section 5.5), whatever the holder's pair. Steps: (a) Dividing by num:  $\Delta L - \kappa^2 L = -(3/\nu)\sigma$ , whose lattice Green's

function is  $G_\kappa(\mathbf{r}) = \int \frac{d^3k}{(2\pi)^3} \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{6-2\sum_a \cos k_a + \kappa^2}$ , far away  $e^{-\kappa r}/(4\pi r)$  per unit source: the level is  $3e^{-r/R}/(4\pi\nu r)$  per unit  $\sigma$  with  $R = 1/\kappa$ , Yukawa's kernel with the massless one's constant, the range the gap's and the strength at the Link the massless one's:  $G_\kappa(1) = G_0(1)(1 - 0.84\kappa + O(\kappa^2))$ , so the strength at one Link is the massless one to one part in  $R$  (and the factor  $1/\nu = 1 + O(\text{gap})$ ). Check: by the integral on a  $400^3$  grid,  $G_0(1) = 0.0857$  (Watson's exact 0.08607),  $G_{1/20}(1) = 0.0822$ , the ratio 0.958 at  $R = 20$  Links;  $G_{1/400}(1)/G_0(1) = 0.9995$ . (b) The coupling. The sign holder's force constant,  $\alpha_{\text{law}}$  (S.26), is the write's half, the kernel's  $3/(4\pi)$  and light's  $\sqrt{3}$ , none of them the holder's own; a gapped holder written and read by the same acts has the same three factors and the kernel above, so it couples at  $\alpha_{\text{law}}(1 - 1/R)$  at one Link,  $1.08\alpha_{\text{law}}$  on the lattice at  $r = 1$ , and  $\alpha_{\text{law}}e^{-r/R}/r$  beyond: every rotation holder Rule3 steps, massless or gapped, abelian or three-line, couples at the charge's  $\alpha$  at the Link. A coupling  $k/(\Gamma E_s)$  itself, the file's  $1/28.33$ , would need the factor  $8\pi/(3\sqrt{3}) = 4.84$  on the kernel, which no gap supplies; a contact strength  $3/\kappa^2$  appears only for  $\kappa > 1$ , a range below one Link, whose quanta would be above the band's top. (c) The numbers. The  $W$ 's range  $\hbar/(m_W c) = 2.45 \times 10^{-3}$  fm and the pion's 1.41 fm are  $1.8 \times 10^8$  and  $1.0 \times 10^{11}$  Links at the Link's bound  $1.4 \times 10^{-26}$  m (S.24), so  $1/R$  is  $6 \times 10^{-9}$  and  $10^{-11}$ : the strength at the Link is the massless one to those parts. A pion-gap holder at its range with the law's coupling binds  $\alpha\hbar c e^{-1/R} = 1.44 \text{ MeV fm} \times 0.368/1.41 \text{ fm} = 0.38 \text{ MeV}$  against the deuteron's 2.22, short by 5.8; a  $W$ -gap holder with the law's  $\alpha$  and nature's  $m_W$  gives  $G_F = \pi\alpha/(\sqrt{2}m_W^2) = 2.5 \times 10^{-6} \text{ GeV}^{-2}$  against nature's  $1.166 \times 10^{-5}$ , short by 4.7, or  $m_W = 37 \text{ GeV}$  from  $G_F$  against 80.4. The nearness of  $1/28.33$  to nature's  $\alpha_3$  at the Link scale and  $\alpha_2$  at  $m_W$ , 0.7 percent apart from each other in nature, is one coincidence at a few percent probability, used for nothing. Check of Yukawa's identity: the range  $R = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  equals the Compton length  $c/\omega_g$  of the holder's own travelling quanta to the gap's first order (S.11), with no input.

**Derivation S.40** (the GHZ correlation under the declared credit, Section 9.4). Inputs: the GHZ family, light's pair with four real lines laid as one event and never summed at a Node, every part laid alike and stepped by the same line, so the four are identical at every detector (the determinism of equal parts, Theorem of Section 9.4); three detectors  $A, B, C$ , each with its setting  $(p, q) = (\cos a, \sin a)$  up to the norm and its pattern, one integer pair per part:  $A(p, p, q, q)$ ,  $B(p, q, p, q)$ ,  $C(p, -q, -q, -p)$ , the  $+$  port reading the part  $k$  as  $e_k(+) = \alpha_k p + \beta_k q$  and the  $-$  port as  $e_k(-) = \alpha_k(-q) + \beta_k p$ ; the credit, the pair's own over three sides:  $J(\pm, \pm, \pm) = (\sum_{k=1}^4 e_k(A) e_k(B) e_k(C))^2$  on equal parts, the common factor of the parts' amplitudes leaving every ratio. Steps: (a) With  $c_a = \cos a$ ,  $s_a = \sin a$  and the same for  $b$  and  $c$ , the  $+$  ports read  $A$  as  $(c_a, c_a, s_a, s_a)$ ,  $B$  as  $(c_b, s_b, c_b, s_b)$  and  $C$  as  $(c_c, -s_c, -s_c, -c_c)$ , so  $\sum_k e_k(A) e_k(B) e_k(C) = c_a c_b c_c - c_a s_b s_c - s_a c_b s_c - s_a s_b c_c = \cos(a + b + c)$ .  $A$   $-$  port replaces  $(p, q)$  by  $(-q, p)$ :  $A$ 's  $-$  port reads  $(-s_a, -s_a, c_a, c_a)$ , and the sum is  $-s_a c_b c_c + s_a s_b s_c - c_a c_b s_c - c_a s_b c_c = -\sin(a + b + c)$ ; by the symmetry of the three patterns every port triple with an even number of  $-$  ports gives  $\pm \cos(a + b + c)$  and with an odd number  $\pm \sin(a + b + c)$  (for instance  $(-, -, +)$  gives  $-\cos$ ,  $(-, -, -)$  gives  $\sin$ ). (b) The eight joint shares are therefore  $\cos^2(a + b + c)$  at the four even triples and  $\sin^2(a + b + c)$  at the four odd, summing to 4 at every setting; the normalised shares are  $\cos^2/4$  and  $\sin^2/4$ . (c)  $E_3$ , the shares signed by the product of the three signs, is  $4 \cdot \cos^2/4 - 4 \cdot \sin^2/4 = \cos 2(a + b + c)$ . A pairwise  $E_{AB}$  sums over  $C$ 's two ports first, which gives  $(\cos^2 + \sin^2)/4 = 1/4$  for each of the four sign pairs of  $A$  and  $B$ , so  $E_{AB} = (1 - 1 - 1 + 1)/4 = 0$ , and the same for the other two pairs; each single marginal is  $1/2$ : no signalling and no correlation in any two, the correlation standing in the three alone. (d) Mermin's combination  $M = E(a, b', c') + E(a', b, c') + E(a', b', c) - E(a, b, c)$  at  $a = b = c = 0$  and  $a' = b' = c' = \pi/4$  is  $\cos \pi + \cos \pi + \cos \pi - \cos 0 = -4$ ,  $|M| = 4$ , where every local assignment of signs to the three ports, and every mixture of them, gives  $|M| \leq 2$  (Mermin 1990). (e) With integer settings  $(p, q)$  the norms  $p^2 + q^2$  multiply every share alike and cancel, so  $x = (1, 0)$  and  $y = (1, 1)$  give the eight shares exactly, 0 and  $1/4$  on the three worlds with one  $x$  ( $E_3 = -1$ ) and  $1/4$  and 0 on  $(x, x, x)$  ( $E_3 = 1$ ),  $M = -4$  exactly. Check: the trigonometric identity  $\cos(a + b + c) = c_a c_b c_c - c_a s_b s_c - s_a c_b s_c - s_a s_b c_c$  and  $\sin(a + b + c) = s_a c_b c_c + c_a s_b c_c + c_a c_b s_c - s_a s_b s_c$ ; the implementation's four checked worlds read the prediction's eight shares,  $E_3 = -1, -1, -1, 1$ , every marginal  $1/2$ , every pairwise  $E$  0 and  $M = -4$  exactly (Section 3.7). Nature's angles: the law's  $E_3 = \cos 2(a + b + c)$  is nature's  $\cos(\theta_A + \theta_B + \theta_C)$  at  $\theta = 2a$ , nature's  $x$  and  $y$  at  $\theta = 0$  and  $\pi/2$  the law's bases  $(1, 0)$  and  $(1, 1)$ , as Bell's angles are (S.10). Status: a theorem under the declared credit, the credit the one assumption (Section 9.4).

**Derivation S.41** (the three predictions by name, Section 10.2). (1) A clock in two places. Inputs: a bound body's beat between two of its modes,  $\Delta = \omega_j - \omega_i$ , scales by the clock  $N = e^{-U}$  at the body's Node (S.31, exactly at long wavelength, to  $(1 - N^2)\omega^2/24$  at finite  $\omega$ ); a record laid on two paths is one record (Section 5.1), and the body's internal state is carried on each path by its own rotation. Steps: on two paths at the contents  $U_1$  and  $U_2$  for  $T$  intervals the internal beats accumulate the phases  $\Delta N_1 T$  and  $\Delta N_2 T$ , differing by  $\Delta(N_1 - N_2)T = \Delta \Delta U T$  to first order (write  $\Omega$  for the beat  $\Delta$ ); for a body prepared in an equal superposition of the two modes the internal states on the two paths overlap as  $|\cos(\Omega \Delta U T/2)|$ , which multiplies the path fringes' visibility. Check: strontium's clock line 429.228 THz,  $\Omega = 2\pi \times 4.29228 \times 10^{14} = 2.697 \times 10^{15} \text{ s}^{-1}$ ;  $\Delta U = gh/c^2 = 9.807 \times 1/(2.998 \times 10^8)^2 = 1.091 \times 10^{-16}$  for one metre;  $T = 1 \text{ s}$ :  $\Omega \Delta U T = 0.294 \text{ rad}$ , the visibility  $\cos(0.147) = 0.989$ ; the correction  $(1 - N^2)\omega^2/24$  is  $10^{-38}$  at nature's gap at the axis bound,  $10^{-19}$  at LHAASO's. The semiclassical expectation (Zych, Costa, Pikovski and Brukner 2011) has

the same magnitude; the law's content is that the one record's beat reads the content at its Node, so the dephasing is the clock's and no new line. (2) The gravity of light. Inputs: the one write: a holder of the content gains the form  $D$  of every record over its wall (Section 3.5); Theorem 3 and S.5: a travelling light record's form summed over its Nodes is its share, its count  $n$  (energy  $nT$ ), and  $D = A^2 \sin^2 \omega$  per Node for a travelling wave as for a standing one; the tension sources the axis parts alone, which enter the Link's factor and not the clock (Section 3.4). Steps: a light packet of count  $n$  writes  $n$  quanta per interval into the massless row as a static body of count  $n$  does, so its far level is  $3n/(4\pi r)$  over  $E_g$  and the clock's well  $U = 3n/(4\pi \Gamma E_g r)$ , Eq. (18)'s  $G_{\text{clock}}$  times  $n$ : a static body falls toward the packet at  $G_{\text{clock}} n/\tau^2$ , that is  $1 \times E/c^2$  for the packet's active mass. In general relativity a pencil of light pulls a static body at  $2 \times E/c^2$  (Tolman, Ehrenfest and Podolsky 1931), the pressure  $p = \rho c^2/3$  along the beam entering  $\rho + 3p/c^2$ ; in the model the stress of the wave,  $T_{aa} = 2 \text{num} b^2 \sin^2 k$  (S.7), enters the Link's factor on its axis and the clock reads the form alone, so the factor is 1: a prediction for a world, no click of nature yet. (3) The black hole's quanta. Inputs: the composed paces (S.3):  $p_0 = \Gamma(1 - 1/\Gamma)^c$ ,  $p_i = p_0^2/\Gamma$  rounded to the unit, every Link coefficient  $R_a = 2 \text{num} p_i^2 q^2/\Gamma^2$ ; Theorem 1, the step a bijection; a collapse ends in Nodes whose content exceeds the frozen content (Section 7.2). Steps:  $p_i$  rounds to 0 where  $\Gamma e^{-2U} < 1/2$ , that is  $U > \frac{1}{2} \ln(2\Gamma)$ , the frozen content  $c_f = (\Gamma/2) \ln(2\Gamma)$  ( $4.70\Gamma$  levels, 28,178 at  $\Gamma = 6,000$ ); there every one of the six coefficients is 0, the Node is cut from its neighbours, no current crosses its Links and no travelling event leaves it; the step is a bijection, so the frozen Nodes' levels are kept exactly and nothing is thermalised or traced out: the region emits nothing,  $T_H = 0$ . Hawking's temperature for one solar mass is  $\hbar c^3/(8\pi G M k_B) = 6.17 \times 10^{-8}$  K. The region's entropy is the count of its kept levels, the width's logarithm per frozen Node, extensive in the frozen volume and not in the area. Check:  $\hbar c^3/(8\pi G M_\odot k_B)$  with  $M_\odot = 1.988 \times 10^{30}$  kg gives  $6.17 \times 10^{-8}$  K;  $\frac{1}{2} \ln(12,000) = 4.696$ .

**Derivation S.42** (what a body writes into the holder of the sign, Section 7.4). Inputs: the continuity identity of S.33: at every Node  $w[W_i(t+1) - W_i(t)]$  equals the net of the Link fluxes  $R_a g$ , so  $\sum_i W_i$  over a body is exactly conserved for a static self-level and a single Node's  $W_i$  changes only by a current through its Links; in particular at one isolated Node,  $z_{t+1} + z_{t-1} = 2 \cos \omega_t z_t$  with any real  $\omega_t$  keeps  $W = \text{Im}(\bar{z}_t z_{t-1})$  exactly ( $W_{t+1} = \text{Im}(2 \cos \omega_t |z_t|^2 - \bar{z}_{t-1} z_t) = W_t$ ). The holder's line at the vacuum's pace,  $\ell_{t+1} - 2\ell_t + \ell_{t-1} = \frac{1}{3} \Delta \ell + s_t$ , with  $s$  the write per interval (the Wronskian over  $E_s T$ ): the wave equation at the speed  $1/\sqrt{3}$  with the retarded solution  $\ell(\mathbf{r}, t) = 3s(t - \sqrt{3}r)/(4\pi r)$  for a point source; a static field has  $D = 0$  and  $F = 0$  and carries no quanta (Section 3.5); the share of a travelling wave of the holder's record is  $A^2 \sin^2 \Omega/T$  quanta per Node (S.5 at  $[1, 1]$ ) moving at  $1/\sqrt{3}$ . Steps: (a) A steady rotation. A body at rest with the standing plane record  $z_i = \phi_i e^{-i\omega_b t}$ ,  $\phi$  real up to one phase, has  $W_i = \phi_i^2 \sin \omega_b$  constant at every Node and the Link quantity  $g_{ij} = \text{Im}(\bar{z}_i z_j) = 0$ : the source is static, the holder's level is its rest, no travelling event is born, no click anywhere. (b) Uniform motion. The source moves uniformly through the lattice below light's speed (the matter band's largest group speed is  $c \cos \omega_0 < c$ , S.21), so its retarded level is the uniformly moving rest, which carries no energy flux to infinity: no light, as for a uniformly moving charge in Maxwell's theory; a body outrunning the holder's waves would write a Cherenkov cone, which no body of the model can. (c) A breathing body. Breathing is a change of the amplitudes  $A_i(t)$  across the body by currents between its Nodes; the total  $\sum_i W_i$  is kept, so the monopole of the source is constant and radiates nothing. The far level of a spherically symmetric source distribution  $\sigma_i(t)$  of constant total  $S$  is, by the retarded integral with the retardation expanded ( $\sigma(t - \sqrt{3}|\mathbf{r} - \mathbf{r}'|) = \sigma(t_r) + \sqrt{3}(\dot{\mathbf{r}} \cdot \mathbf{r}')\dot{\sigma} + \frac{3}{2}(\dot{\mathbf{r}} \cdot \mathbf{r}')^2 \ddot{\sigma} + \dots$ , the first order averaging to 0 over the sphere and  $\langle (\dot{\mathbf{r}} \cdot \mathbf{r}')^2 \rangle = r'^2/3$ ),

$$\ell(r, t) = \frac{3S}{4\pi r} + \frac{3}{4\pi r} \cdot \frac{1}{2} \ddot{Q}(t - \sqrt{3}r), \quad Q(t) = \sum_i \sigma_i(t) r_i^2,$$

so a body of mean square radius  $\langle r^2 \rangle$  breathing as  $A_i(t) = A_i(1 + \delta \cos \Omega t)$  has  $Q = S \langle r^2 \rangle (1 + 2\delta \cos \Omega t)$  to first order and writes the travelling level  $\ell_1 = 3s_{\text{eff}}/(4\pi r)$  at the frequency  $\Omega$  with  $s_{\text{eff}} = \delta \Omega^2 S \langle r^2 \rangle$ : the light born carries the breathing's frequency  $\Omega$  and not the rotation  $\omega_b$ , as Section 7.4 says, with the amplitude  $S \cdot 2\delta$  of Section 7.4 multiplied by  $\Omega^2 \langle r^2 \rangle/2$ , the second moment's factor. The quanta radiated per interval through a far sphere are the share density times the speed times the area,  $4\pi r^2 \cdot \ell_1^2 \sin^2 \Omega/(\sqrt{3}T) = (3\sqrt{3}/4\pi) s_{\text{eff}}^2 \sin^2 \Omega/T$ , of order  $\delta^2 \Omega^6 S^2 \langle r^2 \rangle^2/T$ : the scalar analogue of the quadrupole formula, and a far detector's click rate is that current's share through its front. Section 7.4's three conclusions stand (no light from a steady rotation or a uniform motion, light at the beat  $\Omega$  from a breathing body), the amplitude is to be replaced. Check:  $9/(4\pi\sqrt{3}) = 3\sqrt{3}/(4\pi) = 0.4135$ ; the static kernel  $3/(4\pi r)$  is S.25's; the retarded expansion is the standard multipole one at the speed  $1/\sqrt{3}$ .

**Derivation S.43** (the two-mode line under the rotation, Section 7.4). Inputs: the plane record's step under the holder of the sign (Section 5.5),  $U_t[2 \cos \omega\text{-operator}]$  in the form  $z_{t+1} = U_t(w^{-1}[S z_t + \sum_a R_a(e^{i\theta_a} z_{t+a} + e^{-i\theta_a} z_{t-a})] - U_{t-1} z_{t-1})$  with  $U_t = e^{-i\theta_t}$  the time turn between  $t$  and  $t+1$  and  $\theta_a$  the Link phases, both the holder's level over  $\Gamma$  at the interval's start; light as the holder's own record, a level  $b_L \cos \omega_L t$  at the body with the gradient  $\nabla b_L$  across it; the body's bound modes  $\phi_m$ , real eigenvectors of the symmetric  $\mathbf{M}$  with  $(\mathbf{M}\phi_m)_i = 2w \cos \omega_m \phi_{m,i}$ , orthonormal in the Node weights. Steps: (a) The uniform turn is a gauge. Put  $z_t = e^{-i\Phi_t} u_t$  with  $\Phi_{t+1} = \Phi_t + \theta_t$ . Then  $U_t z_t = e^{-i\Phi_{t+1}} u_t$ ,  $U_t U_{t-1} z_{t-1} = e^{-i\Phi_{t+1}} u_{t-1}$  and  $z_{t+1} = e^{-i\Phi_{t+1}} u_{t+1}$ , so  $u$  obeys the plain line  $u_{t+1} = \mathbf{M} u_t/w - u_{t-1}$  exactly, for any

$\theta_t$ : a time turn uniform over the body changes the record's phase and nothing of its moduli, its form or its modes; at a single Node light read as a turn drives no resonance and no absorption. (b) A constant turn. For  $\theta_t = \theta$  the plane solution  $z = e^{-i\Omega t}$  has  $2\cos(\Omega + \theta) = 2\cos\omega_b$ , so the sense + rotates at  $\omega_b - \theta$  and the sense - at  $-\omega_b - \theta$ : both shift by  $-\theta$ , their magnitudes split by  $2\theta$ , the energy shift following the record's own sense, Section 5.5's line. (c) The Link phases couple the modes. To first order in  $\theta_a$  the Link read adds  $Vz = i \sum_a R_a [\theta_{i,a} z_{i+a} - \theta_{i-a,a} z_{i-a}]$  at the Node  $i$ , an  $i$  times a real antisymmetric matrix, Hermitian. Write  $z_t = \sum_m c_m(t) \phi_m e^{-i\omega_m t}$  with  $c$  slowly varying;  $w(z_{t+1} + z_{t-1}) - \mathbf{M}z_t = -2iw \sum_m \sin\omega_m c'_m \phi_m e^{-i\omega_m t}$ , and projecting  $Vz$  on  $\phi_j$  with  $\theta_{i,a}(t) = \theta_L \epsilon_a(i) \cos\omega_L t$  ( $\epsilon_a$  the light's profile across the body,  $\theta_L = b_L/\Gamma$ ) gives  $\langle \phi_j, V\phi_m \rangle = i\theta_L \cos(\omega_L t) K_{jm}$  with

$$K_{jm} = \sum_{\text{Links } (i,i+a)} R_a \epsilon_a(i) [\phi_j(i)\phi_m(i+a) - \phi_m(i)\phi_j(i+a)], \quad K_{mj} = -K_{jm}, \quad K_{mm} = 0,$$

the inter-mode Link current of S.33's  $g$  taken between two modes. So

$$-2iw \sin\omega_j c'_j = i\theta_L \cos(\omega_L t) \sum_m K_{jm} c_m e^{-i(\omega_m - \omega_j)t};$$

the product carries the frequencies  $\omega_m - \omega_j \pm \omega_L$  and resonates only at  $\omega_L = |\omega_j - \omega_m|$ : the difference line alone, since  $V$  is linear in  $z$  and does not mix the two senses, so the sum line  $\omega_i + \omega_j$  and the one-mode line  $2\omega_b$  of the pace read (S.16) are absent; and  $K_{mm} = 0$  says a Link phase shifts no single mode at first order, the gauge statement of (a) for the Links. (d) Rabi's rate. At  $\omega_L = \omega_j - \omega_m$  the resonant terms are  $c'_j = -\theta_L K_{jm} c_m / (4w \sin\omega_j)$  and  $c'_m = +\theta_L K_{jm} c_j / (4w \sin\omega_m)$ , so  $c''_j = -\Omega_R^2 c_j$  with

$$\Omega_R = \frac{\theta_L |K_{jm}|}{4w \sqrt{\sin\omega_j \sin\omega_m}},$$

a full transfer between the two modes every half period, one transition per click, with S.16's detuning law  $\pm \sqrt{\Omega_R^2 - \Delta^2}/4$  and width  $2\Omega_R$ . (e) The matrix element is the dipole. For a uniform gradient,  $\epsilon_a = 1$ , the identity  $\sum_i x_a(i) [\phi_j(\mathbf{M}\phi_m) - (\mathbf{M}\phi_j)\phi_m]_i = 2w(\cos\omega_m - \cos\omega_j) \langle \phi_j, x_a \phi_m \rangle$ , evaluated with  $\mathbf{M}$ 's hopping and  $x_a(i+a) - x_a(i) = 1$ , gives  $K_{jm} = 2w(\cos\omega_j - \cos\omega_m) d_{jm} = -4w \sin \frac{\omega_j + \omega_m}{2} \sin \frac{\omega_j - \omega_m}{2} d_{jm}$  with  $d_{jm} = \langle \phi_j, x_a \phi_m \rangle$  the dipole: the velocity form equals the length form times the line's frequency, the discrete Ehrenfest relation; for a small line  $\Omega_R = \theta_L \omega_L |d_{jm}|/2$ , nature's  $d \cdot E_0/(2\hbar)$  with the field  $E_0 = \omega_L A_0$  and  $\theta_L$  the phase per Link of the vector potential, the rotating-wave factor  $1/2$  included; a static Link phase ( $\omega_L = 0$ ) mixes the modes without a transition, Stark's shift. Check: the slowly varying method is S.16's;  $K_{mm} = 0$  for every real mode by antisymmetry; the identity in (e) follows from  $\mathbf{M}$ 's hopping terms alone, the  $S$  term cancelling. What rests on it: the body's click of S.14, one transition at the line  $\omega_j - \omega_i$  taking the share  $T(\sin\omega_j - \sin\omega_i)$ , stands under the rotation read as under the pace read, the difference line being the one line under the rotation; the transfer rate is the law's  $\Omega_R$  above with the current's matrix element; the fraction absorbed, with the body's write back, stays open as Section 7.4 says. Status: derived, one derivation; the pace read's one-mode line (S.16) has no analogue under the rotation.

**Derivation S.44** (the two forces between quanta and their ratio between families, Section 10.1). Inputs: the sign holder's coupling between two quanta,  $\alpha_{\text{law}} = (3\sqrt{3}/8\pi) k/(\Gamma E_s)$  (S.26); gravity's level from a source of  $s_g$  wells per interval,  $3s_g/(4\pi r \Gamma E_g)$  far away, Eq. (18) of the paper (S.25), and the clock's rate  $1 - U$  in it (S.2); the quantum of a family, its energy  $T \sin\omega_s$ , its count  $\sin\omega_s$  and its Wronskian  $T/2$ . Steps: (a) The two held rows are one kind of row, the kernel  $3/(4\pi r)$  and the same acts of write and read. The sign holder is written by a quantum's Wronskian,  $1/(2E_s)$  per interval for every family, and the content holder by its form,  $\sin\omega_s/E_g$  per interval, so a quantum of the family  $s$  at  $r$  writes the level  $U(r) = 3 \sin\omega_s/(4\pi r \Gamma E_g)$ . (b) A reader quantum of the family  $s'$  has its rotation redshifted by  $U$ , its energy by  $\delta E = -T \sin\omega_{s'} U$ , so the pull is  $F_g = 3T \sin\omega_s \sin\omega_{s'}/(4\pi \Gamma E_g r^2)$ ; the electric force between two quanta by S.26 (c) is  $F_e = 3kT \cos\omega/(8\pi \Gamma E_s r^2)$ ,  $\cos\omega \rightarrow 1$  at a small gap. In the form of a coupling, with light's  $c = 1/\sqrt{3}$  as in S.26 (b), gravity's between two quanta of one family is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2\omega_s/(\Gamma E_g)$ , the square of the quantum's energy over  $T$ . (c) The ratio at one family:  $F_e/F_g = \alpha_{\text{law}}/\alpha_g(s) = k E_g/(2E_s \sin^2\omega_s)$ , the unit of charge ( $T/2$ ) over the quantum's energy ( $T \sin\omega_s$ ), squared, times the file's two weights  $k$  and  $E_g$  over  $E_s$ . (d) Between two families the weights cancel:  $(F_e/F_g)_e/(F_e/F_g)_p = \sin^2\omega_p/\sin^2\omega_e = (m_p/m_e)^2$  with no number, the masses being the energies  $T \sin\omega_s$  (S.22). Check: nature's  $e^2/(4\pi\epsilon_0 G m_e^2) = 4.17 \times 10^{42}$  and  $e^2/(4\pi\epsilon_0 G m_p^2) = 1.24 \times 10^{36}$ , the ratio  $3.37 \times 10^6 = 1836.15^2$ ; at the LHAASO bound  $\omega_e < 1.1 \times 10^{-9}$ ,  $1/\sin^2\omega_e > 8.3 \times 10^{17}$  and  $k E_g/E_s = 2 \times 4.17 \times 10^{42} \sin^2\omega_e$ , about  $10^{25}$  there and larger at a smaller  $\omega_e$ ; on the board the two rows carry two integers and one kernel, and the  $10^{42}$  appears only when a detector compares the two forces on one quantum, its charge read as its invariant and its mass as its energy.

**Derivation S.45** (the nuclear holder's well: the threshold, the binding at a size, the capture and its reverse, Section 10.3). Inputs: a gapped holder of the content with the reach  $1/\kappa$  (S.11) and a declared weight  $W$ , read by the nucleon

families alone and sourced by them at the same weight; the far level of a source with the reach,  $3W \sin \omega_p e^{-\kappa r} / (4\pi r)$  per quantum (S.25, S.39); the clock's read of a level,  $\omega \rightarrow \omega e^{-x}$  with  $x = L/\Gamma$  (S.2); the matter band's inertia  $m^* = 3 \tan \omega_p$ ,  $3\omega_p$  at a small gap (S.22); the second moment's light (S.42) and the two-mode line under the rotation (S.43). Steps: (a) The well and its threshold. Each quantum reads the other's level into its clock, a potential  $-\omega_p x$  per quantum in rotation units, so the pair's potential is  $V(r) = -g^2 e^{-\kappa r} / r$  with  $g^2 = 3W\omega_p^2 / (2\pi\Gamma)$ , and the relative motion has the inertia  $\mu = m^*/2 = 1.5\omega_p$ . Yukawa's well binds if and only if  $2\mu g^2 / \kappa \geq 1.6798$ , that is  $W \geq 1.173 \Gamma \kappa / \omega_p^3$ : 591 in the implementation's own units ([2, 3] as the nucleon, the binding holder [2400, 2401] at the range 20 Links,  $\Gamma = 6,000$ ), and  $7 \times 10^{10} \Gamma$  at nature's deuteron range 1.41 fm and the LHAASO Link. (b) The binding at a body's size. A body of radius  $R$  Links of quanta of the gap  $\omega_s$  standing in any well of the  $1/r$  kind that fits its size has by the virial the binding fraction  $b = 1/(6\omega_s^2 R^2) = (\lambda_s/R)^2/2$  with  $\lambda_s = 1/(\sqrt{3}\omega_s)$  its Compton length, the kinetic rotation over the rest rotation, whatever the holder's weight: the proton at 0.84 fm  $3.1 \times 10^{-2}$ , the deuteron at 2.13 fm  $4.9 \times 10^{-3}$ , iron at 3.75 fm  $1.6 \times 10^{-3}$ , against nature's  $1.19 \times 10^{-3}$  and  $9.4 \times 10^{-3}$ . (c) The capture. Two slow packets entering the well are a superposition of the bound mode and the continuum; the relative coordinate breathes at the beat  $\Omega$  between them, the proton's charge rides it, so the body writes a dipole  $d = Sr_p$  into the holder of the sign,  $S = k_w/(2E_s)$  and  $r_p = r/2$ , the first order of S.42's expansion since the charge is not symmetric about the centre, and radiates at the beat in the dipole's form, the time level carrying half of Larmor's power in form and the odd lines the rest, their coefficient open; the body's own paces move with the level it writes, so it settles into the bound mode deterministically, the binding leaving as light at the beat; no rate and no settling time is claimed. (d) The reverse. A light record of rotation  $\omega_L \geq B$  at the body drives the bound mode into the continuum by the two-mode line under the rotation (S.43), the matrix element the inter-mode current, the dipole; the bound mode's tail  $e^{-\sqrt{2\mu B} r} / r$  and the continuum's density give the shape  $(\omega_L - B)^{3/2} / \omega_L^3$ , Bethe and Peierls's, peaking at  $2B$ . (e) The size law. A cloud of  $n$  quanta in one well has  $R^* \propto 1/n$  and  $b \propto n^2$ , iron 28 times smaller than the deuteron and bound 800 times harder, against nature's  $A^{1/3}$  and its saturation; the second holder, of shorter range and negative weight, is the saturating declaration. Check:  $1.6798 \times 2\pi/9 = 1.1727$ ; at [2, 3],  $\omega_p = 0.8411$  and  $\kappa = 1/20$ :  $1.173 \times 6,000 \times 0.05/0.595 = 591$ ;  $\lambda_p = 0.210$  fm,  $(0.210/2.13)^2/2 = 4.9 \times 10^{-3}$ . Status: one derivation under the hypothesis, the normalisation of the radiated share for a second; nothing run.

**Derivation S.46** (the atom with the nucleus's angle alone, the record's own level, and the holders the electron does not read, Section 7.3). Inputs: the sign holder's angle of a nucleus of charge  $Z$  at  $r$ ,  $\theta_0 = Z\alpha_{\text{law}}/r$  by the far kernel (S.26); a bound electron record of the matter family read into Schrödinger's slow limit (Section 3.3) with the band's inertia  $m^* = 3 \tan \omega_s$  per interval (S.22) and light's  $c = 1/\sqrt{3}$ ; the one write of every record into the one holder (Section 3.4). Steps: (a) Bohr's form with the nucleus's angle alone. The slow limit is the hydrogen-like problem with the coupling  $Z\alpha$ , the inertia  $m^*$  and the speed  $c$ : the levels  $E_n = -m^*(Z\alpha)^2 c^2 / (2n^2) = -m^*(Z\alpha)^2 / (6n^2)$  per interval, Rydberg's  $1 - 1/n^2$  between them, and the radius  $a_0 = 1/(m^*Z\alpha c) = \sqrt{3}/(m^*Z\alpha)$  Links, which at nature's gap ( $m^* = 3\omega_s$ ) is  $\bar{\lambda}_e/(Z\alpha)$ ; the lines are the beats of two levels (S.43); the electron's rotation is redshifted by the binding,  $(Z\alpha)^2/2$  of its energy at a small gap, the mass defect as the click layer reads a frequency. (b) The orbital moment. The kinetic term of a plane record in a plaquette flux  $\Phi$  (S.33's covector) gives the orbital splitting  $\Phi m_\ell / (2m^*)$ , one Bohr magneton per unit of angular momentum: the orbital Zeeman  $g = 1$ ; the spin term's  $g = 2$  is the dimension-4 family's (Section 10.3). (c) The record's own level. The holder of the sign is one field, and a record cannot tell its own write from the nucleus's, so the bound electron reads the nucleus's angle  $Z\alpha/r$  and its own, of the opposite sense and the strength  $1/Z$  of the nuclear one, spread over its own radius: the Hartree self-term, repulsive and at full strength. With  $x = a_0/a$  for a hydrogen-like trial record the energy over Rydberg's unit is  $x^2 - 2x + (1.25/Z)x$ , whose minimum converges to Bohr's as  $1/Z$ ; at  $Z = 1$  it is  $-0.14$  of Rydberg's at  $a = 2.7a_0$ , and Rydberg's ratios do not hold: a finding under a second derivation, the law's way out, if any, not yet found. So the atom is reflected on the board in Bohr's form for  $Z\alpha \gg \alpha$ , a supporting finding and not a claim of nature's hydrogen. (d) The holders the electron does not read. A nuclear holder the electron reads is excluded by the spectrum: a one-Node proton by the finite-size part of hydrogen's 1S level, 1.1 MHz for nature's radius, which a point nucleus would lack at 500 times its 2.2 kHz uncertainty, and by the form factor, 1 at every momentum transfer where nature's falls; a 1 fm cloud bound by a gapped holder at the binding's content 8/938 by muonic hydrogen, 6.5 meV against the 0.09 to 0.15 meV agreement of the electronic and muonic radii, by 40 to 70; an ungapped holder by the spectrum at nine orders. So the nuclear holders are read by the nucleon families and not by the electron, the selective read of Section 4.1 (the numbers computed by hand from the law's lines, one derivation). Check:  $\alpha^2/2 = 2.66 \times 10^{-5}$ ;  $x^2 - 0.75x$  has its minimum at  $x = 0.375$ ,  $-0.1406$ ; at  $Z = 30$  the self-term is 3 percent of the nuclear one. Status: (a) and (b) derived in form, two derivations; (c) one derivation, the second asked; (d) computed.

**Derivation S.47** (the equivalence principle between families, Section 8 (i)). Inputs: the fall of a free packet of the pair [num, den] in a pace gradient,  $a = c^2 [2\nu/(1 + \nu)] \nabla U$  with  $\nu = \text{num}/\text{den}$  (S.18), the proton and the neutron two families of the gaps  $\omega_p$  and  $\omega_n$  under the families' reading. Steps:  $2\nu/(1 + \nu) = 1 - \tan^2(\omega_s/2)$  exactly, so two families fall apart by  $[\tan^2(\omega_n/2) - \tan^2(\omega_p/2)]$ ,  $(\omega_n^2 - \omega_p^2)/4$  at small gaps. Two bodies differ in their fall by that times the difference of their neutron fractions: beryllium (5/9) against titanium (26/48) differ by 0.0139, and with  $\omega_n^2 - \omega_p^2 = (1838.68^2 - 1836.15^2)\omega_e^2 = 9,297\omega_e^2$  the Eötvös parameter is  $\eta = 0.0139 \times 2,324\omega_e^2 = 32.3\omega_e^2$ , below

$3.9 \times 10^{-17}$  at the LHAASO bound  $\omega_e < 1.1 \times 10^{-9}$ , under nature's  $10^{-15}$ ; the electron against the proton falls apart by  $\omega_p^2/4 = 10^{-12}$  at the same bound, which no click has read (the electron's free fall known to 10 percent). Check:  $(1 - \nu)/(1 + \nu) = \tan^2(\omega/2)$  with  $\nu = \cos \omega$ ;  $2 \times 4.17 \times 10^{42}$  is not used here;  $(1838.68^2 - 1836.15^2) = 9,297$ ;  $9,297/4 \times 0.0139 = 32.3$ . Status: derived, two derivations; clicks.